QUANTUM IAS & PCS

TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

World Geography

PYQ-Based Revision Pointers

Chapters 1 to 20

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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How to Use This Book

Every pointer is a real question. Each pointer is the answer to a UPPSC question asked between 1990 and 2025, written as a fact you can revise in seconds, with the related fact you need for the next variation.

Frequency marks. ■ and a red count (e.g. 4×) mark a fact asked 3 or more times. • and a gold count mark facts asked once or twice. The years follow; "one-liner" means a short starred question in the source with no year given.

Asked Most box. Each chapter opens with the facts UPPSC repeats most. Revise these first.

Diagrams. Flowcharts and diagrams turn sequences (timelines, procedures, processes, river courses) into one picture.

Trap Alerts. The places where the obvious answer is wrong or two facts are commonly swapped.

Last-Minute 10. The ten lines to read on the morning of the exam.

Pairs with the Masterbook. Use this book for revision after the TATTVA PYQ Masterbook, which has the full questions, options and explanations.

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Chapter 1

The Universe, the Solar System and the Sun

Built from 43 UPPSC questions (1990–2019). Focus: Chandrasekhar and black holes, planet records, Venus, Saturn, the Sun's energy and distance.

ASKED MOST · REVISE FIRST

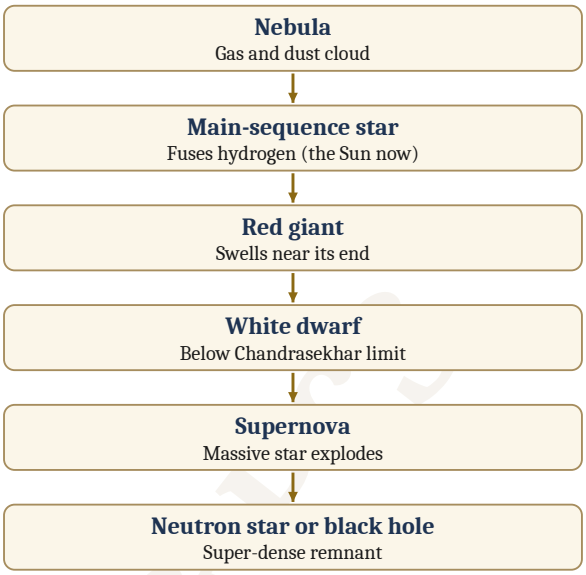
- **Black hole** theory: **S. Chandrasekhar**; Chandrasekhar limit = collapse limit of stars **8×** 1996, 1997, 1997, 2001, 2003, 2015, 2016, 2019
- **Mercury**: nearest the Sun, shortest year, smallest planet now **3×** 1991, 2002, 2004
- **Venus**: Morning Star, Earth's twin; with Mercury has **no moons** **4×** 2007, 2008, 2015, 2016
- **Saturn**: rings, Titan, Cassini (2004) **4×** 1990, 2004, 2006, 2010
- Solar energy = **nuclear fusion** **3×** 1996, 2001, 2006

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

1.1 Universe, Galaxies, Stars

- **Big Bang** explains the origin of the **universe** **1×** 2016
- Milky Way: a **spiral galaxy** **1×** 2001
- Group of stars in a shape: **constellation** **1×** 2013
- Hubble's first detailed image of a star's surface: **Betelgeuse** **1×** 1995
- Star-collapse limit: **Chandrasekhar limit** (1.4 solar masses) **1×** 1997
- Black hole theory: **S. Chandrasekhar**; first information **4×** 1996, 2015, 2019, 2016
- Star-related phenomenon: **black hole** **1×** 1997
- Black hole unseen because even **light cannot escape** (R explains A) **2×** 2001, 2003
- **Supernova**: a **dying star** **1×** 2015
- Not in space: **brittle star** (a sea animal) **1×** 1996
- Not a space term: **byte** **1×** 1997

LIFE OF A STAR



1.2 Planets

- Number of planets: **8** since Pluto became a dwarf planet (2006); 9 then (*) **1×** 1992
- Planets **shine but are non-luminous** (reflect light) **1×** 1992
- Order from Sun: **Earth, Jupiter, Uranus, Pluto** **2×** 2002, 2003
- Between Earth and Sun: **Mercury and Venus** **1×** 2010
- Earth lies between **Venus and Mars** **1×** 2012
- Nearest the Sun: **Mercury**; shortest orbit: **Mercury** **2×** 2004, 2002
- Smallest planet: **Mercury** (Pluto then) **1×** 1991
- Largest: **Jupiter** **1×** 1990
- Wrong pair: slowest planet-**Mars** (it is **Neptune**) **1×** 2011
- Longest revolution among options: **Pluto** (Neptune among planets) **1×** 2004
- No satellites: **Mercury and Venus** **1×** 2016
- Morning Star: **Venus**; Earth's twin: **Venus** **3×** 2015, 2007, 2008
- **Magellan** spacecraft: **Venus** **1×** 1991
- Day and tilt like Earth's: **Mars**; possible life: **Mars** **2×** 2001, 2000
- Jupiter's ring: **silicates** (dust) **1×** 2004
- Rings: **Saturn**; Cassini orbited **Saturn** (2004) **2×** 1990, 2004
- Saturn is **warmer than Neptune** **1×** 2010
- **Titan**: Saturn's largest moon **1×** 2006
- **Nix and Hydra**: moons of **Pluto** **1×** 2010
- H, He, CH4 together: giant planets (no option exact, *) **1×** 2004
- **Asteroids**: between **Mars and Jupiter** **1×** 2008

Planet	Record
Mercury	Nearest, smallest, 88-day year; no moon
Venus	Brightest, hottest; spins backwards; no moon
Mars	Red planet; Earth-like day and tilt

Planet	Record
Jupiter	Largest; fastest spin; Ganymede (largest moon)
Saturn	Rings; least dense; Titan; most moons
Uranus	Rolls on its side
Neptune	Farthest; longest year (165 years)

1.3 The Sun

- Sun's core: **plasma** (no option exact, *) 1× 2001
- Solar energy: **nuclear fusion** 3× 1996, 2001, 2006
- Solar eclipse on **new moon** day; when the **Moon is between Earth and Sun** 2× 1991, 1990
- Earth-Sun distance: **149.6 million km** 1× 2010
- Nearest the Sun (perihelion): **3 January** 1× 2010

TRAP ALERT

- **Hottest = Venus** (not Mercury); **largest moon = Ganymede** (not Titan).
- Pluto options: check if the question is **before 2006**.
- Solar eclipse = **new moon**; lunar eclipse = **full moon**.

UP LINK

No UP item. Indian link: **Aditya-L1** (2023) studies the Sun from the L1 point; the Indian Standard Meridian passes near **Mirzapur**, UP.

LAST-MINUTE 10

1. Black hole, limit = **Chandrasekhar**
2. Supernova = dying star; Milky Way = spiral
3. 8 planets since **2006**
4. Mercury: nearest, smallest; Jupiter: largest
5. Venus: Morning Star, twin, hottest
6. No moons: **Mercury, Venus**
7. Mars: Earth-like day; Saturn: rings, **Titan**
8. Asteroids: **Mars-Jupiter**
9. Sun: **fusion**; 149.6 million km; perihelion **3 Jan**
10. Solar eclipse on new moon

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Chapter 2

The Earth and the Moon

Built from 15 UPPSC questions (1992–2016). Focus: Earth's crust and axis, the Moon's facts, blue moon, lunar missions.

ASKED MOST · REVISE FIRST

- Only one face of the Moon is seen: rotation = revolution period 1× 2006
- **Blue moon**: two full moons in one month 2× 2007, 2009
- Match: **planet-Uranus, satellite-Moon, comet-Halley, spacecraft-Mariner** 2× 1992, 1999

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 The Earth

- Earth's nickname: **Green Planet** (per key) 1× 1999
- Most abundant element in the crust: **oxygen** 1× 2013
- Sunlight reaches Earth in about **8 minutes** 1× 2005
- Axis tilted **23.5°** (66.5° to orbital plane) 1× 2016
- True: northern end of axis = North Pole; axis stays parallel (1 and 3) 1× 2000
- A ball thrown from a satellite keeps **orbiting in the same path** 1× 1995

Crust element	Share
Oxygen	46.6%
Silicon	27.7%
Aluminium	8.1% (most common metal)
Iron	5.0% (most in the whole Earth)

2.2 The Moon

- One face seen because rotation = revolution (R explains A) 1× 2006
- Moon's core: **viscous liquid** per key (now thought solid inner + fluid outer) 1× 2004
- No sound on the Moon: **no atmosphere** 1× 2003
- Half moon: Sun-Earth-Moon angle **90°** 1× 2005
- Match: **planet-Uranus, satellite-Moon, comet-Halley, spacecraft-Mariner** 2× 1992, 1999
- Blue moon: **two full moons in a month** 2× 2007, 2009
- On the Moon an object's **weight decreases** (mass same) 1× 2006
- Earth-Moon distance: about **3,84,400 km** (key 364 thousand, perigee) 1× 2008
- SELENE: **Japan**, first Asian lunar orbiter of its kind 1× 2009

MILESTONES ON THE MOON

Luna 2 (1959)
USSR; first impact

↓

Apollo 11 (1969)
USA; first humans

↓

Chandrayaan-1 (2008)
India; water molecules

↓

Chang'e 4 (2019)
China; far side

↓

Chandrayaan-3 (2023)
India; south pole; 23 Aug = National Space Day

TRAP ALERT

- Most abundant in **crust** = oxygen; in **whole Earth** = iron.
- Weight changes on the Moon; **mass does not**.
- Blue moon is about **two full moons**, not colour.

UP LINK

No UP item. **Chandrayaan-3** landing day, 23 August, is now **National Space Day**.

LAST-MINUTE 10

1. Crust: **oxygen**, silicon, aluminium
2. Light from Sun: **8 min 20 s**
3. Axis tilt **23.5°**
4. Moon: same face, rotation = revolution
5. No air, so no sound on the Moon
6. Half moon: angle **90°**
7. Blue moon = 2 full moons in a month
8. Moon distance **3,84,400 km**
9. **SELENE** = Japan
10. Chandrayaan-3 = south pole, 2023

Chapter 3

Latitudes, Longitudes, Tropics and Seasons

Built from 25 UPPSC questions (1990–2020). Focus: time calculation, Prime Meridian and Date Line, countries on the Equator and Tropics, solstices and equinoxes.

ASKED MOST · REVISE FIRST

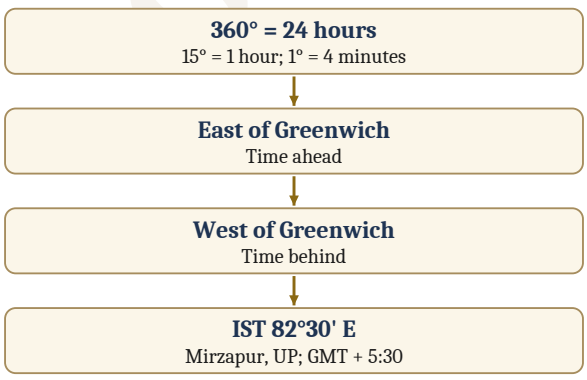
- Time: **15° = 1 hour**; 5 hours ahead = **75° E**; 3 hours ahead = **45° E** 2× 2006, 2008
- Equator crosses **Kenya, Colombia, Gabon, Somalia**; not Rwanda or Equatorial Guinea 4× 2007, 2008, 2010, 2020
- **21 June**: longest day, smallest noon shadow, summer solstice 3× 1992, 2006, 2010
- Equinoxes: **21 March and 23 September** 2× 1994, 2004
- Prime Meridian crosses **France and Algeria** 2× 2008, 2010

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

3.1 Latitude, Longitude, Time

- Equator meets Prime Meridian in the **Atlantic Ocean** (Gulf of Guinea) 1× 2009
- Farthest north of the options: **Rome** 1× 2011
- Noon at Greenwich, 5 pm local: **75° E** 1× 2008
- GMT 3 am, local 6 am: **45° E** 1× 2006
- Great circle with Prime Meridian: **180°** 1× 2000
- First sunrise of the millennium in India: **93°30' E** (Arunachal) 1× 2001
- Prime Meridian crosses **Algeria and France** 2× 2008, 2010
- Date Line 12 hours from Greenwich, at 180° (R explains A) 1× 1992
- Date Line has no minimum distance (per key) 1× 1997
- Ship moving at Earth's spin speed: still **10:00 Monday** 1× 2000
- Shortest path: along a **meridian** (great circle) 1× 1992

TIME FROM LONGITUDE



3.2 Equator and Tropics

- Equator does not pass: **Equatorial Guinea and Rwanda** 1× 2020
- On the Equator: **Colombia and Kenya**; **Kenya** 3× 2010, 2007, 2008
- Equator and both Tropics cross: **Africa** 1× 2004
- Tropic of Cancer does not cross **Iran** 1× 2005
- Tropic of Capricorn does not cross **Bolivia** 1× 1997
- Capricorn crosses **Australia, Namibia, Brazil, Chile** 1× 2003

Line	Key countries
Equator	Gabon, DR Congo, Uganda, Kenya, Somalia, Indonesia, Ecuador, Colombia, Brazil
Tropic of Cancer	Mexico, Egypt, Saudi Arabia, UAE, Oman, India, Bangladesh, Myanmar, China
Tropic of Capricorn	Chile, Argentina, Paraguay, Brazil, Namibia, Botswana, South Africa, Madagascar, Australia

3.3 Solstices and Seasons

- Equinoxes: **21 March and 23 September** 2× 1994, 2004
- Longest day: **21 June**; summer solstice **21 June** 2× 1992, 2010
- Smallest noon shadow: **21 June** 1× 2006
- 12-hour day at summer solstice: **Equator** 1× 1997
- Noon Sun at Tropic of Cancer on winter solstice: **43°** 1× 1997
- Midnight sun: **North Pole** (and Arctic) 1× 1990

TRAP ALERT

- **Iran** is not on the Tropic of Cancer; **Bolivia** not on Capricorn.
- **Equatorial Guinea** (mainland) and **Rwanda** are off the Equator.
- Earth is **nearest the Sun in January**, yet it is northern winter.

UP LINK

IST is based on **82°30' E**, which passes near **Mirzapur** (UP). The Tropic of Cancer crosses 8 Indian states but **not UP**.

LAST-MINUTE 10

1. 15° = 1 hour; 1° = 4 minutes
2. 5 h ahead = **75° E**; 3 h ahead = **45° E**
3. Prime Meridian: **France, Algeria**
4. Date Line **180°**
5. Equator: **Kenya, Colombia**; not Rwanda
6. All three lines: **Africa**
7. Not on Cancer: **Iran**; not on Capricorn: **Bolivia**
8. Equinox **21 Mar, 23 Sep**
9. Solstice **21 June**: longest day
10. IST: 82°30' E near **Mirzapur**

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Chapter 4

Geological History, Volcanoes and Earthquakes

Built from 25 UPPSC questions (2004–2021). Focus: Pangaea and Gondwana, active volcanoes and their countries, Ring of Fire, epicentre and seismograph.

ASKED MOST · REVISE FIRST

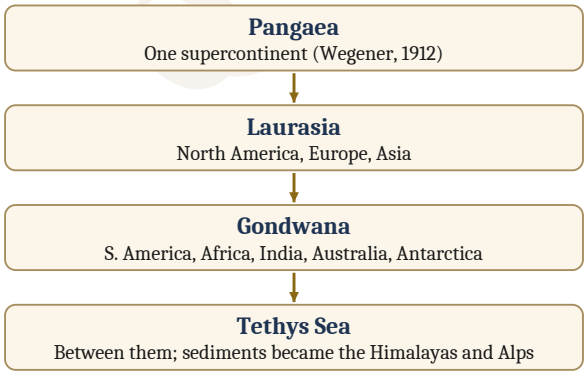
- **Volcano-country matching**: **Etna-Sicily (Italy), Vesuvius-Italy, Erebus-Ross Island, Cotopaxi-Ecuador, Merapi and Krakatoa-Indonesia, Colima and Paricutin-Mexico, Apo-Philippines, Rainier-USA** 4× 2010, 2015, 2016, 2021
- Active volcanoes: **Cotopaxi, Etna, Fuji, Vesuvius**; Aconcagua and Kilimanjaro are not 2× 2007, 2011
- **Mauna Loa, Kilauea** = Hawaii; Kilauea most active 4× 2005, 2006, 2009, 2014
- **Ring of Fire** = earthquakes, volcanoes, **Pacific** 1× 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

4.1 Geological History

- Age of the Earth: **uranium (radiometric) dating** 1× 2006
- Oldest fossils: about **3.5 billion** years (all options wrong, *) 1× 2005
- Original landmass: **Pangaea** 1× 2016
- Not part of Gondwana: **North America** 1× 2016
- Order: **shelled animals, insects, reptiles, mammals** 1× 2018
- Little Ice Age: **1650-1870 AD** 1× 2015
- Son valley formations order: **Sihawal, Patpara, Baghor, Khetaunhi** 1× 2017
- Thar desert: **Pleistocene and recent deposits** 1× 2018

PANGAEA BREAKS UP



4.2 Volcanoes

- Highest volcano among options: **Cotopaxi** (overall: Ojos del Salado) 1× 2015
- **Mauna Loa**: **Hawaii** 2× 2005, 2014
- Most active: **Kilauea** 2× 2006, 2009
- Active: **Cotopaxi, Etna, Fujiyama** (not Aconcagua) 1× 2007
- Active: **Etna and Vesuvius** 1× 2011
- **Kilimanjaro**: **Tanzania** 1× 2009
- "Lighthouse of the Mediterranean": **Stromboli** 1× 2004
- **Etna-Sicily, Vesuvius-Italy, Erebus-Ross Island, Cotopaxi-Ecuador** 1× 2010
- **Kinabalu-Malaysia, Elburz-Iran, Aconcagua-Argentina, Kilimanjaro-Tanzania** 1× 2015
- **Sabancaya-Peru, Etna-Italy, Colima-Mexico, Merapi-Indonesia** 1× 2015
- **Krakatoa**: **Indonesia** 1× 2016
- **Rainier-USA, Etna-Italy, Paricutin-Mexico, Apo-Philippines** 1× 2021

Volcano	Place
Etna / Vesuvius / Stromboli	Sicily / Naples / Aeolian Is. (Italy)
Kilauea, Mauna Loa	Hawaii
Cotopaxi, Chimborazo	Ecuador
Paricutin, Colima, Popocatepetl	Mexico
Krakatoa, Merapi, Tambora	Indonesia
Pinatubo, Taal, Apo	Philippines
Erebus	Ross Island, Antarctica
Barren Island	India (only active)

4.3 Earthquakes

- Ring of Fire: **earthquakes, volcanoes, Pacific** (1, 2, 3) 1× 2008
- Measured on the **Richter scale** 1× 2015
- Recorded by a **seismograph** 1× 2017
- First to record: **epicentre** (surface point above focus) 1× 2017
- NW India quake-prone due to **plate tectonics** 1× 2005

TRAP ALERT

- **Etna** is on **Sicily**; Vesuvius on the mainland.
- **Aconcagua, Kilimanjaro** are volcanic but **not active**.
- Focus = inside the Earth; **epicentre** = surface above it.

UP LINK

The **Son valley** formations (Sihawal, Patpara, Baghor, Khetaunhi) lie in the middle Son basin just south of UP; India's seismic Zone IV covers western UP and the Terai.

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LAST-MINUTE 10

1. **Pangaea** (Wegener); Gondwana + Laurasia
2. North America not in Gondwana
3. Little Ice Age **1650-1870**
4. Kilauea most active; Mauna Loa = Hawaii
5. **Stromboli** = lighthouse of the Mediterranean
6. Etna = Sicily; Erebus = Antarctica
7. Krakatoa, Merapi = Indonesia; Paricutin = Mexico
8. Ring of Fire = **Pacific**
9. Richter scale; seismograph; epicentre
10. Barren Island = India's active volcano

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Chapter 5

Continents, Mountains, Plateaus and Valleys

Built from 42 UPPSC questions (1992–2024). Focus: continent areas, mountain-country matching, mountain types, Death Valley and basins.

ASKED MOST · REVISE FIRST

- **Mountain-country matching**: **Pyrenees-France/Spain, Apennines-Italy, Balkan-Bulgaria, Alps-Switzerland, Atlas and Toubkal-Morocco, Kilimanjaro-Tanzania, Elburz-Iran, Black Forest-Germany** 9× 2004, 2004, 2010, 2012, 2014, 2014, 2020, 2023, 2024
- **Africa** has the **most countries**; largest is **Asia** (key once said Africa) 2× 2014, 2015
- **Andes** = longest range 2× 2008, 2016
- **Death Valley**: California, USA; rift valley; extreme heat 4× 2005, 2009, 2012, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Continents

- Largest continent: **Africa** per key (actually **Asia**; Africa second) 1× 2014
- Descending: **Africa, South America, Europe, Australia** 1× 2015
- Descending: **Asia, Africa, North America, Europe** 1× 2017
- Smallest: **Australia**; two smallest: **Australia and Europe** 2× 1992, 2008
- Most countries: **Africa** (54) 1× 2015
- Most plains by share: **Europe** 1× 2013

5.2 Mountain Ranges

- Longest range: **Andes** 1× 2016
- Length order: **Andes, Rockies, Himalaya, Great Dividing Range** 1× 2008
- **Andes-South America, Altai-Asia, Alps-Europe, Appalachian-North America** 1× 2023
- West to east: **Pontic, Zagros, Hindu Kush, Karakoram** 1× 2024
- Not Tertiary: **Appalachians** (old fold) 1× 2014
- **Alps-folded, Vosges-block, Vindhya-residual, Fuji-volcanic** 1× 2013
- "Sea of mountains": **British Columbia** 1× 2015
- **White Mountains** and **Black Mountain**: **USA** 2× 2001, 2009
- **Allegheny-USA, Cantabrian-Spain, Elburz-Iran, Mackenzie-Canada** 1× 2010
- **Darling Range**: **south-western** coast of Australia 1× 2020
- India-Myanmar ranges: none of the options (Patkai, Naga, Lushai/Arakan) 1× 1998
- **Black Forest**: **Germany**; along the **Rhine** 3× 2004, 2012, 2014
- **Titlis**: **Switzerland** 1× 2004
- In Italy: **Apennines** 1× 2014
- **Spain-France-Pyrenees, Switzerland-Alps, Bulgaria-Balkan, Italy-Apennine** 1× 2024
- **Atlas-Morocco, Kilimanjaro-Tanzania, Chimborazo-Ecuador, Pyrenees-France/Spain** 1× 2020
- **Kilimanjaro-Tanzania, Toubkal-Morocco, Stanley-Uganda, Hoggar-Algeria** 1× 2023
- **Drakensberg**: **South Africa** 2× 2007, 2008

Type	Examples
Young fold	Himalaya, Alps, Andes, Rockies, Atlas
Old fold	Appalachians, Urals, Aravalli
Block	Vosges, Black Forest, Sierra Nevada
Volcanic	Fuji, Kilimanjaro, Cotopaxi
Residual	Vindhya, Nilgiri, Aravalli

5.3 Plateaus

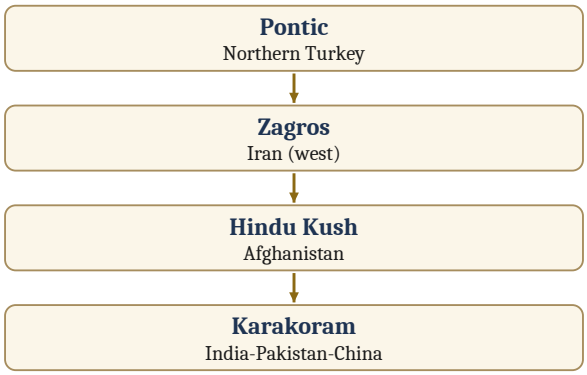
- Tibetan Plateau average height: about **5 km** 1× 2013
- Largest plateau among options: **Colorado** (per key) 1× 2017
- Telegraphic Plateau: part of the **North Atlantic Ridge** 1× 2019

5.4 Valleys and Basins

- **Death Valley**: a **rift valley**; "Devil's Golf Course", **USA**; known for **heat**; in **California** 4× 2012, 2014, 2005, 2009
- **Panjshir Valley**: **Afghanistan** 1× 2002
- **Silicon Valley**: **California** 1× 2002
- **Taylor Valley**: **Antarctica** 1× 1998
- **Great Artesian Basin**: **Australia** 2× 2002, 2003
- **Great Basin**: **USA** 1× 2004
- **Valley of the Kings**: **Egypt** 1× 1998
- **Blind valley**: **karst** region 1× 2013

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WEST TO EAST ACROSS WEST ASIA



TRAP ALERT

- **Great Basin** = USA; **Great Artesian Basin** = Australia.
- **Darling Range** = WA coast; **Darling River** = NSW.
- **Appalachians** are old fold, not Tertiary.

UP LINK

UP links: the **Vindhyan** hills (residual mountains) reach Mirzapur and Sonbhadra; UP lies on the Indo-Gangetic **plain**, built from Himalayan alluvium.

LAST-MINUTE 10

1. Asia largest; **Africa** most countries; Australia smallest
2. **Andes** longest; then Rockies
3. Pyrenees = France-Spain; Apennines = Italy; Balkan = Bulgaria
4. Black Forest = Germany (Rhine); Titlis = Switzerland
5. Atlas, Toubkal = Morocco; Drakensberg = South Africa
6. Elburz = Iran; Mackenzie = Canada; Allegheny = USA
7. Appalachians = old fold; Vosges = block; Vindhya = residual
8. Death Valley = rift, hottest, California
9. Great Artesian Basin = **Australia**
10. Valley of the Kings = **Egypt**

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Chapter 6

Deserts and Grasslands

Built from 31 UPPSC questions (1993–2025). Focus: desert-country matching, Atacama and Kalahari, temperate vs tropical grasslands, Pampas.

ASKED MOST · REVISE FIRST

- **Pampas** = **Argentina** (temperate grassland, main farm belt) 5× 2005, 2012, 2016, 2017, 2018
- **Grassland matching: Llanos-Venezuela/S. America, Steppe-Eurasia, Prairies-N. America, Veld-South Africa, Downs-Australia, Puszta-Hungary** 6× 2002, 2010, 2014, 2015, 2016, 2017
- **Savanna** = tropical, mostly **Africa**, "land of big game" 5× 1993, 2011, 2012, 2013, 2019
- **Kalahari** = Botswana, Namibia; Southern Hemisphere 3× 2008, 2010, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Definition and Location

- Desert: rainfall under **25 cm** a year 1× 2009
- Hot deserts on **western margins at 15°-30°**; trade winds lose moisture moving west (R explains A) 1× 2019
- Deserts can supply solar energy; they get a year's human energy use in 6 hours of sun (R explains A) 1× 2013

6.2 Deserts of the World

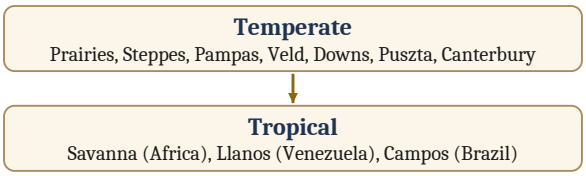
- Biggest desert: **Sahara** (largest hot desert) 1× 1994
- Smallest to largest: **Great Sandy, Gobi, Arabian, Sahara** 1× 2023
- Most populated desert: **Thar**; largest in South Asia: **Thar** 2× 2012, 1996
- **Gobi**: **Mongolia and China** only 2× 1993, 2025
- **Taklamakan**: **China** 1× 2008
- Wrong: **Gibson-Brazil** (Gibson is in **Australia**) 2× 2016, 2018
- **Karakum**: **Turkmenistan** 1× 2022
- **Dasht-e-Lut**: **Iran** 1× 2015
- Not true: Atacama lies in **southern** Chile (it is **northern**) 1× 2006
- Driest place: **Atacama** 1× 2009
- Temperate desert: **Patagonian** 1× 2011
- **Kalahari**: **Botswana and Namibia**; in the **Southern Hemisphere** 3× 2008, 2010, 2015

Desert	Country
Sahara	North Africa
Arabian (Rub al Khali)	Saudi Arabia
Gobi	Mongolia, China
Taklamakan	China (Xinjiang)
Karakum / Kyzylkum	Turkmenistan / Uzbekistan-Kazakhstan
Dasht-e-Lut	Iran
Thar	India, Pakistan
Kalahari / Namib	Botswana, Namibia / Namibia
Atacama	Northern Chile (driest)
Patagonian	Argentina (temperate)
Gibson, Great Sandy, Great Victoria	Australia

6.3 Grasslands

- Wrong: **Veld-Australia** (Veld = South Africa) 1× 2015
- **Llanos-South America, Steppe-Europe/Eurasia, Prairies-North America, Veld-Africa** 1× 2016
- **Pampas-Argentina, Puszta-Hungary, Veld-South Africa, Downs-Australia** 1× 2017
- Wrong: **Downs-tropical** (Downs are **temperate**) 1× 2002
- Not temperate: **Savanna** 2× 2012, 2013
- Eurasian temperate grassland: **Steppes** 1× 2014
- Savanna most widespread in **Africa** 2× 1993, 2011
- "Land of big game": **tropical savanna** 1× 2019
- Treeless South American plains: **Pampas**; in **Argentina** 3× 2005, 2012, 2018
- Pampas main farm region: fertile soil, temperate climate (R explains A) 1× 2016
- Correct: **Llanos-Venezuela** 1× 2010

GRASSLANDS BY TYPE



TRAP ALERT

- **Gibson** = Australia; **Atacama** = northern Chile.
- **Downs** and **Veld** are temperate; **Llanos** and **Campos** are tropical.
- Largest desert overall = **Antarctica**; largest hot = **Sahara**.

UP LINK

No UP item. Indian link: the **Thar**, the most densely populated desert, is fed by the Indira Gandhi Canal, while UP's Bundelkhand faces semi-arid conditions.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Desert: under **25 cm** rain
2. Hot deserts on **west coasts**, 15°-30°
3. Sahara largest hot; Thar most populated
4. Gobi = Mongolia, China; Karakum = Turkmenistan
5. Gibson = **Australia**; Lut = **Iran**
6. Atacama = northern Chile, driest
7. Patagonian = temperate desert
8. Kalahari = Botswana, Namibia
9. Pampas = Argentina; Puszta = Hungary; Downs = Australia
10. Savanna = Africa, tropical; Llanos = Venezuela

QUANTUM IAS & PCS

Chapter 7

Countries and their Borders

Built from 42 UPPSC questions (1994–2023). Focus: areas and coastlines, regional groups, landlocked countries, dependent territories, boundary lines.

ASKED MOST · REVISE FIRST

- **Landlocked**: **Laos** (only one in SE Asia), **Bolivia, Chad**, Afghanistan, Hungary, Switzerland; **Liberia** and **Azerbaijan** (per key) are not 7× 2001, 2005, 2009, 2010, 2013, 2015, 2022
- **North America** has **no** landlocked country 1× 2005
- Boundary lines: **Radcliffe-India/Pakistan, Durand-Afghanistan/Pakistan, Maginot-France/Germany, 38th parallel-Koreas, Rio Grande-USA/Mexico** 5× 1994, 2005, 2008, 2014, 2015
- Longest land border with India: **Bangladesh** 1× 2013

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

7.1 Size and Coastline

- Largest country: **Russia** (not in options, *) 1× 1996
- Ascending area: **Argentina, India, Australia, Brazil** 1× 2012
- Not larger than India: **France** 1× 2013
- Longest coastline: **Canada** 1× 2015
- Longest border with India: **Bangladesh** 1× 2013

7.2 Regions and Neighbours

- Not in the Arabian Peninsula: **Syria** (only 4) 1× 2023
- Not in the Middle East: none clearly (Turkiye least wrong, *) 1× 2021
- Israel borders **Lebanon, Syria, Jordan, Egypt** 1× 2017
- Gaza borders **Egypt and Israel** (both) 1× 2023
- Horn of Africa: **Somalia, Ethiopia, Djibouti**; not **Sudan** 2× 1999, 2009
- Not Scandinavian: **Finland** 1× 2001
- No Baltic coast (per key): **Sweden and Norway** option (Norway has none) 1× 2021
- Not Central American: **Sierra Leone** 1× 2014
- Not in Africa: **Guyana** 1× 2017
- Not on Africa's west coast: **Botswana** 1× 2016
- Zambia borders all four, Botswana by 150 m (*) 1× 2013
- Afghanistan does not border **Russia** 2× 2002, 2012

7.3 South-East Asia and Islands

- Indonesia has no land border with **Brunei** 1× 2021
- Indonesia: over 15,000 islands, over 150 active volcanoes: key says neither 1× 2021
- **Borneo** divided among **three** countries (key also says volcanic) 1× 2022
- Galapagos: **volcanic** (true); part of Peru (false; **Ecuador**) 1× 2021
- **Australia-Christmas Island, Denmark-Greenland, France-Martinique, Spain-Santa Cruz** 1× 2000
- In America but politically European: **Greenland** 1× 2002
- **Aleutian-USA, Bear Island-Norway, Greenland-Denmark, Franz Josef Land-Russia** 1× 2014

7.4 Landlocked and Transcontinental

- Not landlocked: **Liberia** 1× 2005
- Landlocked in Africa: **Chad** 1× 2010
- Only landlocked in SE Asia: **Laos** 2× 2001, 2015
- Landlocked: **Bolivia** 1× 2022
- No landlocked country: **North America** 1× 2005
- Landlocked: **Afghanistan, Hungary, Switzerland** (not Malaysia) 1× 2009
- Not landlocked (Caspian coast): **Azerbaijan** per key 1× 2013
- Landlocked: **Armenia, Botswana, Ethiopia**; Montenegro is coastal (*) 1× 2016
- South Sudan: **landlocked**, main river **White Nile** (I and III); capital Juba 1× 2010
- Transcontinental: **Georgia and Turkey** 1× 2009
- Zimbabwe was **Southern Rhodesia** 1× 2004

Continent	Landlocked examples
Africa (16, most)	Ethiopia, Chad, Niger, Mali, Zambia, Zimbabwe, Botswana, Uganda
Asia	Afghanistan, Nepal, Bhutan, Laos, Mongolia, Central Asian states
Europe	Switzerland, Austria, Hungary, Czechia, Serbia
South America	Bolivia, Paraguay
North America, Australia	None

7.5 Boundary Lines

- Wrong: **Durand Line-Bangladesh/India** (it is **Afghanistan-Pakistan**) 1× 2015
- **Radcliffe Line**: **India and Pakistan** 1× 2014
- **Maginot Line**: **France-Germany** 1× 1994
- Rivers as borders: **Rio Grande** and Rhine (two right, *) 2× 2002, 2003
- **Rio Grande**: **Mexico and USA** 1× 2008
- Koreas divided at **38° N** 1× 2005

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FAMOUS LINES

Radcliffe 1947
India-Pakistan, India-Bangladesh

↓

Durand 1893
Afghanistan-Pakistan

↓

McMahon 1914
India-China (east)

↓

Maginot 1930s
French wall facing Germany

↓

38th parallel
North-South Korea

↓

49th parallel
USA-Canada

TRAP ALERT

- **Guyana** = South America; **Guinea** = Africa.
- **Finland** is Nordic, not Scandinavian.
- Uzbekistan and Liechtenstein are **doubly landlocked**.

UP LINK

UP borders **Nepal** (open border, 1,751 km for India overall), with border districts such as Maharajganj, Siddharthnagar, Bahraich, Lakhimpur Kheri and Pilibhit.

LAST-MINUTE 10

1. Russia largest; India 7th; Canada longest coast
2. Bangladesh = longest border with India
3. Horn of Africa: Somalia, Ethiopia, Eritrea, Djibouti
4. Finland not Scandinavian; Guyana not African
5. Laos = only landlocked in SE Asia
6. Bolivia, Paraguay = landlocked in S. America
7. North America: no landlocked country
8. Greenland = Denmark; Christmas Island = Australia
9. Durand = Afghanistan-Pakistan; Radcliffe = India-Pakistan
10. 38° N = Koreas; Rio Grande = USA-Mexico

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Chapter 8

Capital Cities, Towns and Geographical Sobriquets

Built from 50 UPPSC questions (1990–2024). Focus: capital matching by region, largest city vs capital, Central Asian and African capitals, nicknames.

ASKED MOST · REVISE FIRST

- **Capital matching** is the format in almost every question here **11×** 1992, 1995, 1997, 2003, 2009, 2009, 2011, 2015, 2016, 2019, 2023
- **Central Asia**: **Tashkent, Dushanbe, Bishkek, Ashgabat, Astana**; Bishkek is **not** Tajikistan's **4×** 2007, 2009, 2011, 2019
- **Africa** wrong pairs: **Nigeria-Rabat, Cape Verde-Bamako, Khartoum-Ethiopia** **3×** 2007, 2020, 2024
- **Finland**: Helsinki; "land of a thousand lakes"; Oslo is **Norway's** **4×** 1993, 2001, 2012, 2015

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

8.1 Europe

- Not capitals: **Rotterdam, Zurich** (Riga, Zagreb are) **1×** 2023
- All pairs correct; key took **Berlin-Germany** (*) **1×** 2008
- **Bulgaria-Sofia, Georgia-Tbilisi, Iceland-Reykjavik, Norway-Oslo** **1×** 2003
- **Helsinki**: **Finland**; **Finland-Oslo** is wrong **2×** 1993, 2001
- Wrong: **Christchurch-New Zealand** (capital **Wellington**) **1×** 2010
- Not a national capital: **Bonn** **1×** 2006

8.2 Asia

- **Myanmar-Yangon (then; now Nay Pyi Taw), Cambodia-Phnom Penh, Vietnam-Hanoi, Laos-Vientiane** **1×** 2016
- Wrong: **Khartoum-Ethiopia** (Khartoum = Sudan; Ethiopia = Addis Ababa) **1×** 2024
- Wrong: **Bishkek-Tajikistan** (Bishkek = Kyrgyzstan) **1×** 2007
- **Uzbekistan-Tashkent, Tajikistan-Dushanbe, Kyrgyzstan-Bishkek, Turkmenistan-Ashgabat** **1×** 2019
- **Azerbaijan-Baku, Gambia-Banjul, Kazakhstan-Astana, Latvia-Riga** **1×** 2011
- Correct: **Lebanon-Beirut** **1×** 2010
- Wrong: **China-Shanghai** (Beijing) **1×** 2005
- **Sri Lanka-Colombo, Maldives-Male, Burma-Rangoon, Zimbabwe-Harare, Mauritius-Port Louis** **1×** 1992
- **Palestine-Ramallah, Kosovo-Pristina, Turkish Cyprus-Nicosia, Taiwan-Taipei** **1×** 2015
- Not in the Sunni Triangle: **Basra** **1×** 2022
- Not in Afghanistan: **Fergana, Samarkand** (Uzbekistan) **1×** 2023
- Highest of the options: **Lhasa** **1×** 1993
- Wrong: **Baikonur-Ukraine** (it is in **Kazakhstan**) **1×** 1992

Country	Capital
Kazakhstan	Astana
Uzbekistan / Tajikistan	Tashkent / Dushanbe
Kyrgyzstan / Turkmenistan	Bishkek / Ashgabat
Azerbaijan / Georgia / Armenia	Baku / Tbilisi / Yerevan
Myanmar	Nay Pyi Taw (Yangon earlier)
Laos / Cambodia / Vietnam	Vientiane / Phnom Penh / Hanoi

8.3 Africa

- Wrong: **Nigeria-Rabat** (Nigeria = Abuja; Rabat = Morocco) **1×** 2007
- **Windhoek-Namibia, Accra-Ghana, Nairobi-Kenya, Lusaka-Zambia** **1×** 2009
- **Burundi-Bujumbura (Gitega now), Kazakhstan-Astana, Laos-Vientiane, Mali-Bamako** **1×** 2009
- **Kampala-Uganda, Kigali-Rwanda, Kinshasa-DR Congo, Khartoum-Sudan** **1×** 2023
- **Brazil-Brasilia, Cuba-Havana, Kenya-Nairobi, Zambia-Lusaka** **1×** 1995
- **Australia-Canberra, Tajikistan-Dushanbe, Tanzania-Dodoma, Switzerland-Bern** **1×** 1997
- Wrong: **Cape Verde-Bamako** (Praia; Bamako = Mali) **1×** 2020
- Wrong: **Maldives-Malabo** (Male; Malabo = Equatorial Guinea) **1×** 2017

8.4 Americas and Oceania

- Correct: **Cyprus-Nicosia** **1×** 2012
- Wrong: **Dominica-Santo Domingo** (Roseau; Santo Domingo = Dominican Republic) **1×** 2021
- **NSW-Sydney, Queensland-Brisbane, Victoria-Melbourne, Western Australia-Perth** **1×** 2021

8.5 Towns

- **Pisa** (leaning tower): **Italy** **1×** 1993
- Mega cities on coasts as gateways to ocean routes (R explains A) **1×** 2005
- Not in Silicon Valley: **Santa Rosa** **1×** 2013
- Wrong: **Tasmania-Uganda** (Tasmania = Australia) **1×** 1990
- South of the Arctic Circle: **Iceland** **1×** 2013

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- **Sun City**: **South Africa** 1× 2011
- **Davos-Switzerland, Barcelona-Spain, Auckland-New Zealand, Kandy-Sri Lanka** 1× 2014

LARGEST CITY IS NOT THE CAPITAL



8.6 Sobriquets

- Continent of birds: **South America** 1× 1994
- Blue Grass State: **Kentucky** 1× 2004
- Gateway to the West (Turkey): **Istanbul** 2× 2005, 2010
- Land of the rising sun: **Japan** 1× 1991
- Garden of lakes: **Finland**; thousand lakes: **Finland** 2× 2012, 2015
- City of the Golden Gate: **San Francisco** 1× 2013
- Manchester of the East: **Osaka** 1× 2014
- Pearl of Siberia: **Lake Baikal** 1× 2015
- Land of morning calm: **Korea** 1× 2015
- Oil capital of Europe: **Aberdeen** 1× 2017
- Land of Promise: **Mindanao** 1× 2021

Nickname	Place
Land of the Midnight Sun	Norway
Land of the Thunder Dragon	Bhutan
Land of White Elephants	Thailand
Land of a Thousand Hills	Rwanda
Sugar Bowl / Pearl of the Antilles	Cuba
Playground of Europe	Switzerland
Emerald Isle	Ireland
Big Apple / Windy City	New York / Chicago

TRAP ALERT

- **Dominica** (Roseau) vs **Dominican Republic** (Santo Domingo).
- **Malabo** = Equatorial Guinea; **Bamako** = Mali; **Rabat** = Morocco.
- **Baikonur** cosmodrome is in **Kazakhstan**.

UP LINK

UP's own sobriquets: **Kanpur** = Manchester of the East (Indian usage; UPPSC keys use Osaka), **Varanasi** = oldest living city, **Agra** = city of the Taj, **Lucknow** = city of nawabs.

LAST-MINUTE 10

1. Tashkent, Dushanbe, **Bishkek** (Kyrgyzstan), Ashgabat, Astana
2. Oslo = Norway; Helsinki = Finland
3. Nicosia = Cyprus; Ramallah = Palestine; Pristina = Kosovo
4. Abuja = Nigeria; Rabat = Morocco; Bamako = Mali
5. Khartoum = Sudan; Addis Ababa = Ethiopia
6. Kampala, Kigali, Kinshasa, Dodoma, Windhoek, Accra
7. Baikonur = Kazakhstan; Basra not in Sunni Triangle
8. Osaka = Manchester of the East; Aberdeen = oil capital
9. Finland = thousand lakes; Korea = morning calm
10. Kentucky = Blue Grass; Mindanao = Land of Promise

Chapter 9

Oceans, Seas and Ocean Currents

Built from 35 UPPSC questions (1991–2025). Focus: seas and their neighbours, warm and cold currents by ocean, salinity, tides, deepest points.

ASKED MOST · REVISE FIRST

- **Benguela** = cold current of the **South Atlantic**; **Agulhas** = warm, **Indian Ocean** 6× 2005, 2014, 2017, 2020, 2023, 2025
- Deepest points: **Pacific-Mariana (Challenger), Atlantic-Puerto Rico, Indian-Java (Sunda), Arctic-Molloy/Eurasian Basin** 3× 1991, 2017, 2022
- Cold currents: **Humboldt (Peru)**, California; warm: **Brazil**, Agulhas 4× 1995, 1998, 2017, 2025
- Salinity highest near the **Tropics**; most of it is **sodium chloride** 2× 2000, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

9.1 The World's Water

- **Datum line**: the sea-level reference for heights and depths 2× 1991, 1994
- Water stores, descending: **oceans, ice and glaciers, groundwater, rivers** 1× 2025
- Most fresh water: **glaciers and ice** 1× 2015
- Gravitational soil water tension: **below 1/3 atmosphere** 1× 2010

9.2 Oceans and Seas

- Indian Ocean: Greeks' **Erythraean Sea**; spans **both sides of the Equator** (both) 1× 2025
- **Red Sea**: an **axial trough** (rift) 1× 2017
- **Baku**: on the **Caspian Sea** 1× 2019
- West to east: **Mediterranean, Black, Caspian, Aral** 1× 2016
- **Aral Sea**: **Kazakhstan and Uzbekistan** 1× 2012
- Not on the Black Sea: **Syria** 1× 2021
- Sea between Italy, Sicily, Sardinia, Corsica: **Tyrrhenian** 1× 2010
- Not on the Mediterranean: **Chad** 1× 2010
- Deepest of the options (per key): **South China Sea** 1× 2015
- No coastline: **Sargasso Sea**; known for **seaweed (sargassum)** 2× 2016, 2014

9.3 Ocean Currents

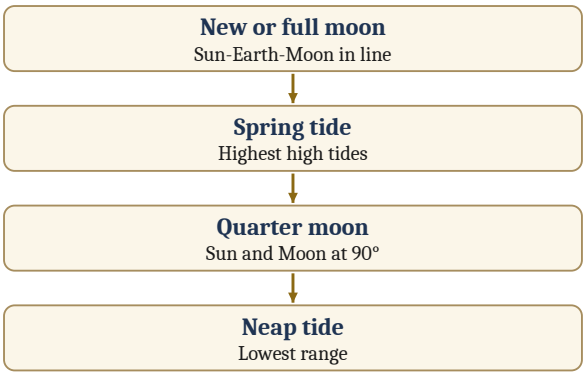
- Indian Ocean current: **Agulhas** 1× 2020
- Not cold: **Agulhas** 1× 2017
- Not in the Indian Ocean: **Benguela** 1× 2014
- Atlantic current: **Benguela** 1× 2023
- Cold, South Atlantic: **Benguela** 1× 2005
- Not in the South Atlantic loop: **Canary** (North Atlantic) 1× 2000
- Not cold: **Agulhas and Brazil** (1 and 2) 1× 2025
- Cold: **Peruvian (Humboldt)** 1× 1995
- Wrong: **Humboldt-North Pacific** (it is **South Pacific**) 1× 1998
- Warm: **Brazil Current** 1× 2017

Ocean	Warm	Cold
North Atlantic	Gulf Stream, N. Atlantic Drift	Labrador, Canary
South Atlantic	Brazil	Benguela, Falkland
North Pacific	Kuroshio	Oyashio, California
South Pacific	East Australian	Humboldt (Peru)
Indian	Agulhas, Mozambique	West Australian

9.4 Salinity and Tides

- Main salt in sea water: **sodium chloride** 1× 2000
- Highest salinity (per key): **Lake Van** (Dead Sea is actually saltier) 1× 1995
- Saltiest sea: Red Sea among seas; Dead Sea overall (*) 2× 1998, 2012
- Salinity is highest near the **Tropics of Cancer and Capricorn** 1× 2023
- Salinity gradient: **halocline** 1× 2015
- Tides: **gravity, centripetal and centrifugal forces** 1× 1991
- **Spring tide**: Sun, Earth, Moon in a **straight line** 1× 1999

SPRING AND NEAP TIDES



9.5 Ocean Deeps

- Deepest trench: **Challenger Deep** (Mariana) 1× 1991
- **Pacific-Mariana, Atlantic-Puerto Rico, Indian-Java, Arctic-Eurasian Basin** 1× 2017

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- **Pacific-Mariana, Arctic-Molloy, Indian-Sunda, Atlantic-Puerto Rico** 1× 2022

TRAP ALERT

- **Benguela** (Atlantic) vs **Agulhas** (Indian), both near South Africa.
- **Canary** is North Atlantic; **Humboldt** is South Pacific.
- Warm + cold meet = fog and fish (Grand Banks, Japan).

UP LINK

No UP item. India link: the **Deep Ocean Mission** and **Samudrayaan** (Matsya-6000) target the Indian Ocean floor.

LAST-MINUTE 10

1. Oceans 96.5%; fresh water mostly in **ice**
2. Red Sea = rift; Sargasso = no coast
3. Baku = Caspian; Aral = Kazakhstan-Uzbekistan
4. Tyrrhenian = Italy-Sicily-Sardinia-Corsica
5. Benguela = cold, S. Atlantic; Agulhas = warm, Indian
6. Humboldt = cold, S. Pacific; Brazil = warm
7. Canary not in the South Atlantic loop
8. Salinity: NaCl; highest near the Tropics; halocline
9. Spring tide: Sun-Earth-Moon in a line
10. Challenger Deep = deepest; Puerto Rico = Atlantic; Java = Indian

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Chapter 10

Rivers of the World

Built from 33 UPPSC questions (1992–2025). Focus: Congo and Limpopo, Nile, Asian rivers and their outflows, Rhine and Mississippi, city-river matching.

ASKED MOST · REVISE FIRST

- **City-river matching**: Berlin-Spree, Rome-Tiber, Paris-Seine, London-Thames, Hamburg-Elbe, Antwerp-Scheldt, Rotterdam-Rhine, Vienna-Danube, Kinshasa and Brazzaville-Congo, Asuncion-Paraguay, Basra-Shatt al-Arab, Niamey-Niger 13× 1998, 1999, 2002, 2004, 2004, 2009, 2015, 2015, 2016, 2016, 2017, 2019, 2025
- **Congo (Zaire)** crosses the Equator twice; **Limpopo** crosses the Tropic of Capricorn twice 2× 2010, 2015
- **Mississippi**: source **Lake Itasca**; **bird's-foot** delta 2× 2015, 2018
- **Rhine**: rift valley; **Rotterdam** at its mouth 3× 1998, 2021, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Africa

- Crosses the Equator twice: **Zaire (Congo)** 1× 2015
- Crosses the Tropic of Capricorn twice: **Limpopo** 1× 2010
- Not in the Nile basin: **Chad** 1× 2013
- **Blue Nile** rises in **Lake Tana** 1× 2016
- Egypt "gift of the Nile", an oasis in the Sahara (R explains A) 1× 2005

THE NILE



10.2 Asia and Europe

- Largest river of Asia: **Yangtze** 1× 2016
- Flows south: **Salween** 1× 2010
- Wrong: **Mekong-south-west** (it flows south/south-east) 1× 2024
- **Lena-Arctic, Amur-Pacific, Tigris-Persian Gulf, Mahi-Arabian Sea** 1× 2016
- Mekong rises in **Tibet** (true); delta in southern Cambodia (false, it is **Vietnam**) 1× 2021
- **Volga** falls into the **Caspian Sea** 1× 1992
- Rhine statements: only 2 correct; **Rotterdam at its mouth** (only 2) 2× 2021, 2023
- Rift-valley river: **Rhine** 1× 1998
- Wrong: **Rhone-Spain** (Rhone = France/Switzerland) 1× 2008
- Best radial drainage: **Sri Lanka**; **Mahaweli Ganga** = Sri Lanka 2× 2008, 2013
- **Chisapani Gorge**: **Nepal** (Karnali) 1× 2015
- **Nobi and Kanto** plains: **Japan** 1× 2021

10.3 Americas

- **Lake Itasca**: source of the **Mississippi** 1× 2015
- Bird's-foot delta: **Mississippi** 1× 2018
- USA-Mexico border river: **Rio Grande** (key also Colorado) 1× 2023

10.4 Cities and Rivers

- Wrong: **New Orleans-Missouri** (Mississippi) and **Hyderabad-Paleru** (Musi) 1× 2025
- **Shatt al-Arab-Basra, Paraguay-Asuncion, Niger-Niamey, Danube-Vienna** 1× 2016
- **Paris-Seine, Kinshasa-Congo, Bangkok-Chao Phraya, Asuncion-Paraguay** 1× 2019
- Wrong: **Sydney-Seine** 1× 2015
- **Berlin-Spree, Lahore-Ravi, New York-Hudson, Rome-Tiber** 2× 1999, 2002
- Wrong: **Vienna-Volga** (Danube) 3× 1998, 2004, 2016
- Correct: **Hamburg-Elbe** 1× 2004
- **Potomac-USA, Nile-Sudan, Tigris-Iraq, Thames-England** 1× 2003
- Wrong: **Lisbon-Tiber** (Tagus) 1× 2009
- **London-Thames, Antwerp-Scheldt, Rotterdam-Rhine, Hamburg-Elbe** 1× 2015
- **Khartoum-Nile, Brazzaville-Congo, Rotterdam-Rhine, Paris-Seine** 1× 2017

City	River
London / Paris / Rome	Thames / Seine / Tiber
Berlin / Hamburg	Spree / Elbe
Vienna, Budapest, Belgrade	Danube
Lisbon / Warsaw	Tagus / Vistula
Kinshasa, Brazzaville	Congo
Bamako, Niamey	Niger

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City	River
Baghdad / Basra	Tigris / Shatt al-Arab
Bangkok	Chao Phraya
New York / Washington	Hudson / Potomac
Asuncion	Paraguay

TRAP ALERT

- **Vienna** is on the Danube, not the Volga; **Lisbon** on the Tagus, not the Tiber.
- Mekong delta is in **Vietnam**, not Cambodia.
- **Bird's foot** = Mississippi; **arcuate** = Nile, Ganga.

UP LINK

UP link: India's **Ganga** forms an arcuate delta (with the Brahmaputra); the 2025 question used **Hyderabad-Musi** as a home trap. Varanasi is UP's Ganga city, as Vienna is the Danube's.

LAST-MINUTE 10

1. Congo crosses Equator twice; Limpopo crosses Capricorn twice
2. Blue Nile from **Lake Tana**; meets White Nile at **Khartoum**
3. Yangtze = Asia's largest; Volga → Caspian
4. Lena → Arctic; Amur → Pacific; Tigris → Persian Gulf
5. Rhine: rift valley; Rotterdam at mouth
6. Mississippi: Itasca; bird's-foot delta
7. Rio Grande = USA-Mexico border
8. Sri Lanka = radial drainage (Mahaweli)
9. Kinshasa, Brazzaville = Congo
10. Berlin-Spree; Hamburg-Elbe; Antwerp-Scheldt

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Chapter 11

Islands, Lakes and Waterfalls

Built from 33 UPPSC questions (1990–2023). Focus: largest islands, Japan and Indonesia island order, Pacific island groups, lake records, Great Lakes cities, waterfalls.

ASKED MOST · REVISE FIRST

- Lake records: **Caspian** largest, **Baikal** oldest and deepest, **Tanganyika** second deepest and longest **3×** 2008, 2014, 2015
- **Lake Michigan** lies **entirely in the USA** **2×** 2015, 2017
- Japan north to south: **Hokkaido, Honshu, Shikoku, Kyushu**; Honshu largest **3×** 1991, 2017, 2019
- Lake-country wrong pairs: **Onega-Canada, Reindeer-Russia, Five Flower-Mongolia, Malawi-Asia** **4×** 2014, 2015, 2017, 2023

■ asked 3 or more times · asked once or twice **4×** times asked, followed by the exam years

11.1 Islands

- Largest island: **Greenland** **1×** 1993
- Largest of options: **New Zealand's South Island** **1×** 2015
- Correct: **Nuuk** (Greenland) is about 3 m above sea level **1×** 2012
- Divided among three countries: **Borneo** **1×** 2010
- South Indian Ocean: Chagos, Mauritius, Seychelles all fit (*) **1×** 2004
- Not correct: **Falklands in the Indian Ocean** (they are South Atlantic) **1×** 1993
- Socotra: belongs to **Yemen** (not Oman); UNESCO site 2008 (only 2) **1×** 2021
- Salt-pan island: **Saint-Martin** **1×** 2017
- **Macau** returned to China in **1999** **1×** 1990

11.2 Island Groups

- Largest island of Japan: **Honshu** **1×** 1991
- North to south: **Hokkaido, Honshu, Shikoku, Kyushu** **2×** 2017, 2019
- Indonesia west to east: **Sumatra, Java, Bali, Lombok** **1×** 2021
- Melanesia: **Solomon Islands** **1×** 2012
- Not Polynesia: **Tinian** (Micronesia) **1×** 2021
- Not in the Caribbean: **Madeira** (Atlantic, Portugal) **1×** 2021
- "Sheep Islands": **Faroe Islands** **1×** 2015

Pacific group	Islands
Melanesia	New Guinea, Solomons, Vanuatu, Fiji, New Caledonia
Micronesia	Guam, Marshalls, Kiribati, Tinian, Palau, Nauru
Polynesia	Hawaii, Samoa, Tonga, Tuvalu, Tahiti, New Zealand

11.3 Lakes

- Largest lake by area and volume: **Caspian Sea** **1×** 2008
- Oldest and deepest: **Baikal** **1×** 2015
- Second deepest and longest: **Tanganyika** **1×** 2014
- By area: **Superior, Victoria, Great Bear, Baikal**; key order 4, 3, 2, 1 **1×** 2016
- Wrong: world's tallest waterfall-**Tugela** (it is **Angel**) **1×** 2015
- Lake Victoria statements: neither correct **1×** 2021
- Wrong: **Malawi-Asia** (Africa) **1×** 2017
- Wholly in the USA: **Michigan** **2×** 2015, 2017
- On Lake Superior: **Duluth** **1×** 2014
- **Erie-Detroit, Michigan-Gary, Ontario-Hamilton, Superior-Duluth** **1×** 2011
- Wrong: **Onega-Canada** (Russia) **1×** 2023
- **Balkhash**: **Kazakhstan** **1×** 2021
- Wrong: **Reindeer Lake-Russia** (Canada) **1×** 2014
- Wrong: **Five Flower Lake-Mongolia** (China, Jiuzhaigou) **1×** 2015
- World Heritage lake: **Plitvice** (Croatia) **1×** 2014

GREAT LAKES, WEST TO EAST

Superior
Largest; Duluth; Soo Canal to Huron

↓

Michigan
Only one wholly in USA; Chicago, Gary

↓

Huron
Joined to Michigan at Mackinac

↓

Erie
Shallowest; Cleveland, Buffalo; near Detroit

↓

Ontario
Smallest; Toronto, Hamilton; to St. Lawrence

11.4 Waterfalls

- In Venezuela: **Angel Falls** (highest) **1×** 2014
- **Victoria Falls**: on the **Zambezi** (true); Zambia-Mozambique border (false; **Zambia-Zimbabwe**) **1×** 2023

QUANTUM IAS & PCS

TRAP ALERT

- **Socotra** = Yemen; **Madeira** = Portugal (Atlantic).
- **Victoria Falls** = Zambia-Zimbabwe; **Lake Victoria** = Uganda-Kenya-Tanzania.
- Highest fall = **Angel**; Tugela is second.

UP LINK

No UP item. UP has no natural great lakes; its large water bodies are reservoirs such as **Rihand (Govind Ballabh Pant Sagar)** in Sonbhadra.

LAST-MINUTE 10

1. Greenland largest island; Borneo = 3 countries
2. Japan N-S: Hokkaido, Honshu, Shikoku, Kyushu
3. Indonesia W-E: Sumatra, Java, Bali, Lombok
4. Tinian = Micronesia; Solomons = Melanesia
5. Caspian largest lake; Baikal oldest, deepest
6. Tanganyika: longest, 2nd deepest
7. Michigan wholly in USA; Duluth on Superior
8. Balkhash = Kazakhstan; Onega = Russia
9. Angel = highest fall (Venezuela)
10. Victoria Falls = Zambezi, Zambia-Zimbabwe

QUANTUM IAS & PCS

Chapter 12

Straits, Gulfs, Canals and Dams

Built from 24 UPPSC questions (1991–2025). Focus: straits and what they separate, the Suez Canal, Soo and Ludwig canals, dam-river matching.

ASKED MOST · REVISE FIRST

- **Suez Canal** joins the **Red Sea and Mediterranean**; ends at **Port Said and Suez**; cut India-Europe route by about **7,000 km** 8× 1992, 1992, 1995, 2005, 2010, 2013, 2018, 2025
- Straits: **Gibraltar** (Europe-Africa), **Bering** (Asia-N. America), **Bass** (Tasmania), **Hormuz** (Persian Gulf-Gulf of Oman), **Dover** (Channel Tunnel) 6× 1999, 2004, 2005, 2007, 2010, 2023
- Dams: **Hoover-Colorado, Kariba-Zambezi, Aswan-Nile, Panchet Hill-Damodar** 3× 2013, 2015, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 Straits

- Europe from Africa: **Gibraltar** 1× 1999
- Asia from North America: **Bering Strait** 1× 2004
- Tasmania from Australia: **Bass Strait** 1× 2023
- **Hormuz**: between the **Persian Gulf and Gulf of Oman** 1× 2005
- UK-France tunnel: **Strait of Dover**; Dover joins the **English Channel and North Sea** 2× 2007, 2010
- **Lombok Strait**: between **Bali and Lombok** 1× 2017
- **Ten Degree Channel**: **Andaman from Nicobar** 1× 2001

Strait	Separates
Gibraltar	Spain and Morocco
Bosphorus, Dardanelles	Europe and Asia (Turkiye)
Bab-el-Mandeb	Yemen and Djibouti
Hormuz	Iran and Oman
Malacca	Malaysia and Sumatra
Bering	Russia and Alaska
Palk	India and Sri Lanka
Ten Degree Channel	Little Andaman and Car Nicobar

12.2 Gulfs

- Persian Gulf does not border **Oman** (main coast on Gulf of Oman) 2× 2007, 2008
- Correct: **Gulf of Tonkin-Vietnam** 1× 2003

12.3 Canals

- Suez joins the **Red Sea and Mediterranean**; (2 and 3) 2× 1992, 2025
- Ports at the ends: **Port Said and Suez** 1× 2013
- Suez shortened India-West distance; joins Mediterranean and Red Sea (R explains A) 1× 1992
- Largest navigation canal: **Suez** 3× 1995, 2005, 2010
- India-Europe saving: about **7,000 km** 1× 2018
- Lakes north to south: **Manzala, Timsah, Great Bitter, Little Bitter** 1× 2019
- **Soo Canal**: **Superior and Huron** 1× 1991
- Inland waterway in southern Germany: **Ludwig Canal** 1× 2017
- Longest sea bridge then: **Jiaozhou Bay** (Hong Kong-Zhuhai-Macau now) 1× 2009

THE SUEZ CANAL, NORTH TO SOUTH



12.4 Dams

- **Colorado-Hoover, Damodar-Panchet Hill, Nile-Aswan, Zambezi-Kariba** 1× 2013
- **Kariba Dam**: **Zambezi** 1× 2016
- Wrong: **Hoover-Tennessee** (Hoover is on the **Colorado**) 1× 2015
- China's first Brahmaputra dam (2010): **Zangmu** 1× 2013

Dam	River	Country
Three Gorges	Yangtze	China
Hoover	Colorado	USA
Aswan High / GERD	Nile / Blue Nile	Egypt / Ethiopia
Kariba, Cahora Bassa	Zambezi	Zambia-Zimbabwe, Mozambique
Itaipu	Parana	Brazil-Paraguay
Zangmu, Medog	Yarlung Tsangpo	China

TRAP ALERT

- **Hoover** = Colorado; **TVA** dams = Tennessee.
- **Soo** = Superior-Huron; **Welland** = Erie-Ontario.
- Suez has **no locks**; Panama uses locks.

UP LINK

UP link: India's **Damodar Valley Corporation** copied the US **Tennessee Valley Authority**; UP's own big dams include **Rihand** (Sonbhadra) and **Matatila** (Betwa, Lalitpur).

LAST-MINUTE 10

1. Gibraltar = Europe-Africa; Bering = Asia-America
2. Bass = Tasmania; Hormuz = Persian Gulf-Gulf of Oman
3. Dover = Channel Tunnel; Lombok = Bali-Lombok
4. Ten Degree Channel = Andaman-Nicobar
5. Suez: Red Sea-Mediterranean; Port Said, Suez; 1869
6. Suez saves ~7,000 km; lakes Manzala-Timsah-Bitter
7. Soo = Superior-Huron; Ludwig = southern Germany
8. Hoover = Colorado; Kariba = Zambezi
9. Aswan = Nile; Panchet Hill = Damodar
10. Zangmu = China's Brahmaputra dam

QUANTUM IAS & PCS

Chapter 13

The Atmosphere

Built from 18 UPPSC questions (1993–2024). Focus: nitrogen, layers of the atmosphere, ozone and ionosphere, albedo, horse latitudes, Coriolis force, jet streams, clouds.

ASKED MOST · REVISE FIRST

- ■ **Ozone layer** lies in the **stratosphere** 4× 1997, 2002, 2003, 2005
- ■ **Ionosphere** reflects **radio waves**; aurora 4× 2002, 2005, 2013, one-liner
- • **Nitrogen** = largest share of air (78%) 2× 1993, 2017
- • Lowest layer: **troposphere** (weather) 2× 2002, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

13.1 Composition and Layers

- Largest share: **nitrogen**; permanent gas: **nitrogen** 2× 1993, 2017
- Ascending order: **troposphere, stratosphere, mesosphere, ionosphere** (key gave descending, a) 1× 2005
- Lowest layer: **troposphere** 1× 2016
- ■ Ozone layer: **stratosphere** 3× 1997, 2003, 2005
- Deflects radio waves: **ionosphere**; reflects wireless signals 2× 2005, 2013
- Northern Lights: **ionosphere** (thermosphere) 1× one-liner
- Correct: **troposphere-weather, stratosphere-ozone, ionosphere-radio waves**; not mesosphere-aurora (1, 2, 3) 1× 2002

LAYERS, GROUND UP

Troposphere
0-18 km; weather; temperature falls

↓

Stratosphere
Up to 50 km; ozone; jets fly low in it

↓

Mesosphere
50-80 km; coldest; meteors burn

↓

Thermosphere / Ionosphere
80-400+ km; radio reflection; aurora

↓

Exosphere
Outer fringe

13.2 Heat and Pressure

- Air heated mainly **from the Earth's surface** (long-wave); conversion happens at the surface (R explains A) 1× 1997
- **Albedo**: **reflecting power** 1× 2019
- Not true: maximum-temperature zone lies **on the Equator** (it is around 10-20° N, the thermal equator) 1× 2002
- Humid summer heat: **muggy** weather 1× 2002
- **Horse latitudes**: about **30°** 1× 2008
- Air pressure lowest in **summer** 1× 2014

Belt	Latitude
Equatorial low (doldrums)	0-5°
Subtropical high (horse latitudes)	30-35°
Subpolar low	60-65°
Polar high	90°

13.3 Winds, Jets, Clouds

- Leftward deflection in the Southern Hemisphere: **Coriolis force** 1× 2023
- Jet streams are **westerly** (A false); speed 300-500 km/h (R true) 1× 2024
- Highest clouds (of options): **cirrostratus** 1× 2009

TRAP ALERT

- Key used "ascending" for a top-down list; upward order is **4, 3, 2, 1**.
- Jet streams are **westerlies**, not easterlies (except the tropical easterly jet over India).
- Aurora and radio reflection = **ionosphere**, not mesosphere.

UP LINK

UP sits under the **subtropical westerly jet** in winter, which steers **western disturbances** bringing winter rain to western UP.

LAST-MINUTE 10

1. Air: N2 78%, O2 21%, Ar 0.93%
2. Lowest = troposphere (weather)
3. Ozone = **stratosphere**
4. Radio waves, aurora = **ionosphere**
5. Coldest layer = mesosphere
6. Albedo = reflecting power
7. Horse latitudes = **30°**
8. Coriolis: right in N, **left in S**
9. Jet streams: westerly, 300-500 km/h
10. Cirrus family = highest clouds

Chapter 14

Cyclones, Tornadoes and Local Winds

Built from 18 UPPSC questions (1992–2021). Focus: names of tropical cyclones by region, tornado and hurricane scales, local winds and their countries.

ASKED MOST · REVISE FIRST

- Cyclone names: **Willy-willy-Australia, Hurricane-USA, Typhoon-China, Baguio-Philippines, Cyclone-India** 5× 2001, 2005, 2014, 2019, 2020
- Local wind wrong pairs: **Mistral-Australia, Shamal-Austria, Bora-Poland, Blizzard-Chile, Sirocco-France** 5× 2005, 2013, 2016, 2019, 2021
- Tornadoes: **USA**, "Tornado Alley" in the **Mississippi plains**; **Fujita scale** 3× 2005, 2009, 2009

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Cyclones and Tornadoes

- **Australia-Willy-willy, China-Typhoon, India-Cyclone, USA-Hurricane** 1× 2005
- **Willy-willy-Australia, Hurricane-USA, Typhoon-China, Baguio-Philippines** 3× 2014, 2019, 2020
- Willy-willy: cyclone of **NW Australia** 1× 2001
- Tornado origin (key): **Caribbean Sea** 1× 1996
- Tornadoes associated with the **USA** 1× 2009
- **Tornado Alley**: **Mississippi plains** 1× 2005
- Tornado intensity: **Fujita scale** 1× 2009
- Hurricane damage: **Saffir-Simpson scale** 1× 2016
- Did not hit India: **Nargis** (Myanmar, 2008) 1× 2013
- Global warming raises frequency and severity of all listed extremes 1× 2018

Region	Name
Atlantic, Caribbean (USA)	Hurricane
NW Pacific (China, Japan)	Typhoon
Philippines	Baguio
NW Australia	Willy-willy
Indian Ocean	Cyclone

14.2 Local Winds

- Winds to the Mediterranean: westerlies (winter) or trades (summer) (*) 1× 1992
- **Foehn**: **Switzerland** (Alps) 1× 2010
- Wrong: **Bora-Poland** (Adriatic) 1× 2016
- Wrong: **Blizzard-Chile** (Canada, USA) 1× 2005
- "Blood rain" (Sirocco): **Italy** 1× 1998
- Wrong: **Sirocco-France** only (Bora-Italy, Blizzard-Canada right) 1× 2013
- Wrong: **Mistral-Australia** (France) 1× 2019
- Wrong: **Shamal-Austria** (Iraq, Gulf) 1× 2021

WARM VS COLD LOCAL WINDS

Warm, descending
Foehn (Alps), Chinook (Rockies), Santa Ana (California)

↓

Hot, desert
Sirocco (Sahara to Italy), Khamsin (Egypt), Harmattan (W. Africa), Shamal (Iraq), Loo (N. India)

↓

Cold
Mistral (Rhone, France), Bora (Adriatic), Blizzard (Canada), Purga (Siberia)

TRAP ALERT

- **Mistral** = Rhone valley, France; **Bora** = Adriatic.
- **Sirocco** blows from the Sahara **to Italy**; red dust = "blood rain".
- **Fujita** = tornadoes; **Saffir-Simpson** = hurricanes; **Richter** = earthquakes.

UP LINK

UP's own local wind is the **Loo**, the hot dry westerly of May-June; pre-monsoon **dust storms** and **Norwesters** strike eastern India.

LAST-MINUTE 10

1. Willy-willy = Australia; Baguio = Philippines
2. Typhoon = China/Japan; Hurricane = USA
3. Tornado Alley = Mississippi plains
4. Fujita = tornado; Saffir-Simpson = hurricane
5. Nargis hit **Myanmar**
6. Foehn = Switzerland; Chinook = Rockies
7. Sirocco = blood rain, Italy
8. Mistral = France; Bora = Adriatic
9. Shamal = Iraq/Gulf; Blizzard = Canada
10. Loo = north India

Chapter 15

World Climate and Forests

Built from 37 UPPSC questions (1996–2025). Focus: forest types and trees, Mediterranean vegetation names, climate-region matching, Mediterranean and West European climates, records.

ASKED MOST · REVISE FIRST

- **Mediterranean climate**: rain in **winter** from the **westerlies**; dry summer 4× 2004, 2004, 2022, 2024
- Forest-tree matching: **monsoon-teak and sal, equatorial-mahogany, Mediterranean-olive, coniferous-pine and spruce** 2× 2013, 2021
- Mediterranean scrub names: **maquis, chaparral (California), fynbos (S. Africa), matorral (Chile)** 1× 2019
- Climate-region matching: **equatorial-Congo, Mediterranean-California, monsoon-Bangladesh, savanna-Sudan** 3× 1999, 2002, 2003
- Records: wettest **Mawsynram**, coldest **Verkhoyansk** 4× 1997, 1999, 2001, 2004

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Forests

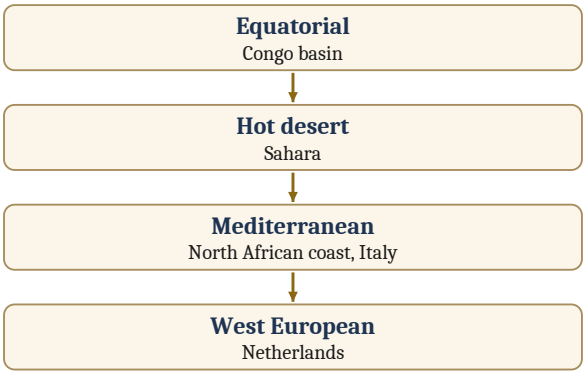
- Highest forest share (of options): **Japan** 1× 2010
- Constitution requires **60%** forest cover: **Bhutan** 1× 2016
- Largest and densest forests: **Brazil** 1× 2004
- Tropical rainforest: **Congo Basin** 1× 1996
- Dense hot forest in Africa: **Ivory Coast** 1× 1998
- **Selva**: **broadleaf evergreen** 1× 2019
- Brazil: rainforests are **selvas**; fertile **terra roxa** (both) 1× 2025
- Soil erosion a big problem in **tropical rainforests** 1× 1998
- **Taiga**: **temperate (cold)** region 1× 2016
- No conifers in **Amazonia** 1× 2001
- **Monsoon-teak and sal, equatorial-mahogany and rosewood, Mediterranean-plum and olive, coniferous-pine and spruce** 1× 2013
- Not Mediterranean: **teak** 1× 2021
- **Maquis-Mediterranean coast, fynbos-southern Africa, chaparral-California, matorral-Chile** 1× 2019
- **Epiphytes-equatorial, acacia-Sahara, baobab-savanna, cedars-Mediterranean** 1× 2019

Forest	Trees
Equatorial (selva)	Mahogany, rosewood, ebony, rubber
Tropical monsoon	Teak, sal, bamboo
Savanna	Acacia, baobab
Mediterranean	Olive, cork oak, cedar, citrus
Taiga	Pine, spruce, fir, larch

15.2 Climate Types

- Main factor: **latitude** 1× 2010
- Rain depends on **humidity** 1× 2017
- Largest share of land: **arid regions** 1× 2002
- All climate zones: **Asia** 1× 2016
- Not equatorial: **Ethiopia** (highland) 1× 2013
- Equator suits plants not people; small annual range (both true, R not the explanation) 1× 2010
- **California-Mediterranean, West Australia-hot desert, Bangladesh-monsoon, Siberia-cold temperate** 1× 1999
- **Equatorial-Congo, Mediterranean-California, monsoon-Bangladesh, savanna-Sudan** 2× 2002, 2003
- Interior Australia is desert (true); northern Australia temperate (false, it is **tropical**) 1× 2025
- Linked to shifting pressure belts: **Mediterranean and monsoon** 1× 2010
- Zaire to Netherlands: **equatorial, hot desert, Mediterranean, West European** 1× 2002

ZAIRE TO THE NETHERLANDS



15.3 Mediterranean and West European

- Correct: **wet winter, dry summer-Italy** 1× 2004
- Winter rain because of the **westerlies** (R explains A) 1× 2004
- Winter rain (true); summer under dry trades (true, R not the explanation) 1× 2022
- Winter rain; belts shift south in winter (both) 1× 2024
- NW European ports open all year; SW winds all year (both true, R not the explanation; warm North Atlantic Drift) 1× 2002
- Western Europe: rain in **all months**, in the **westerlies** (both) 1× 2024

QUANTUM IAS & PCS

15.4 Records and El Nino

- Wettest place: **Mawsynram** 2× 1997, 2001
- Coldest place (key): **Verkhoyansk** 2× 1999, 2004
- **El Nino** develops in the **Pacific** 1× 2015
- El Nino statements: only 1 correct 1× 2023
- **Hekistotherms**: plants of **very low temperature** 1× 2013
- "Vegetation is the true index of climate": **Koppen** 1× 2018

TRAP ALERT

- **Teak** is a monsoon tree, never Mediterranean.
- Northern Australia is **tropical**, not temperate.
- Open winter ports of NW Europe owe more to the **North Atlantic Drift** than to winds.

UP LINK

UP has **tropical monsoon (deciduous)** forests of **sal** in the Terai and teak-dry deciduous in Bundelkhand; Dudhwa's sal forests are the State's best.

LAST-MINUTE 10

1. Bhutan: **60%** forest by Constitution
2. Selva = broadleaf evergreen; Amazon, Congo
3. Taiga = pine, spruce; no conifers in Amazonia
4. Maquis, chaparral, fynbos, matorral = Mediterranean scrub
5. Teak and sal = monsoon forest
6. Mediterranean: winter rain from **westerlies**
7. West Europe: rain all year; westerlies
8. Asia has every climate zone
9. Mawsynram wettest; Verkhoyansk coldest (key)
10. Koppen: vegetation indexes climate

QUANTUM IAS & PCS

Chapter 16

Major Tribes of the World

Built from 23 UPPSC questions (1996–2025). Focus: tribe-homeland matching, Bushmen, Maasai, Bedouin, Semang, Arctic peoples.

ASKED MOST · REVISE FIRST

- ■ **Bushmen (San)** = **Kalahari**, Botswana, Namibia **8×** 1996, 2005, 2008, 2008, 2009, 2012, 2016, 2025
- ■ **Maasai** = **Kenya** (East Africa); never West Africa or Zambia **5×** 2005, 2006, 2009, 2009, 2025
- ■ **Bedouin** = Arabia; **Eskimo/Inuit** = Canada; **Pygmies** = Congo basin **5×** 2001, 2005, 2009, 2016, 2025
- • **Semang** = **Malaysia**, not Indonesia **2×** 2002, 2010

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

16.1 Tribes and Homelands

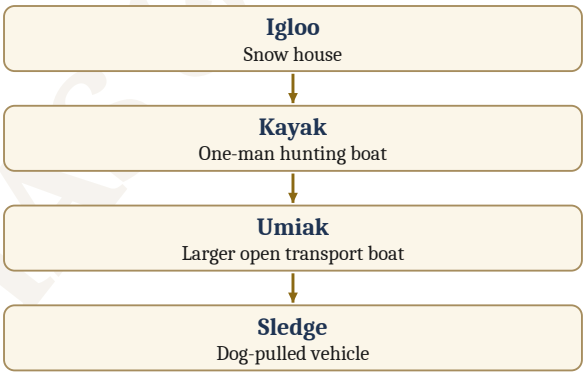
- Correct: **Eskimo-Canada** **1×** 2001
- Correct: **Bushman-Kalahari** **1×** 2012
- Wrong: **Maasai-West Africa** only **1×** 2025
- **Eskimo-Canada, Maasai-Kenya, Bedouin-Saudi Arabia, Bushman-Botswana** **1×** 2009
- **Khirghiz-Central Asia, Bushman-Kalahari, Ainu-Japan, Bedouin-Arab** **1×** 2016
- **Bushman-Namibia, Bedouin-Saudi Arabia, Pygmies-Congo, Maasai-Kenya** **1×** 2005
- **Jews-Israel, Teda-Libya, Beja-Egypt, Lur-Iran** **1×** 2020
- All pairs acceptable (Bushmen also in South Africa) (*) **1×** 1996
- Bushmen: **Kalahari**; correct: **Bushmen-Botswana** **2×** 2008, 2008
- Wrong: **Maasai-Western Africa**; **Maasai-Zambia** **2×** 2006, 2009
- Wrong: **Bantu-Sahara** **2×** 2010, 2012
- Wrong: **Khirghiz-Sudan** **1×** 2019
- Wrong: **Congo basin-Negrito** (Pygmies) **1×** 2004

People	Homeland
Inuit (Eskimo)	Alaska, Canada, Greenland
Sami (Lapps)	Scandinavia
Kirghiz	Central Asia
Bedouin	Arabian desert
Maasai	Kenya, Tanzania
Pygmies	Congo basin
Bushmen (San)	Kalahari
Semang	Malaysia
Ainu	Japan (Hokkaido)
Maori	New Zealand

16.2 Asia and the Arctic

- Semang live in **Malaysia**; **Semang-Indonesia** is wrong **2×** 2002, 2010
- Wrong: **Alaska-Koryak** (Koryak = Kamchatka, Russia) **1×** 2021
- Inuit not found in **Sweden** **1×** 2009
- **Lapps (Sami)**: **Scandinavia** **1×** 2010
- **Kayak-narrow hunting boat, "Inuit" (igloo)-snow house, Umiak-transport boat, Sledge-dog vehicle** **1×** 2014

ARCTIC WAY OF LIFE



16.3 Ways of Life

- Not pastoral nomads: **Pygmies** (hunter-gatherers) **1×** 2005
- Native North Americans: **Mongoloid** **1×** 2015

TRAP ALERT

- **Maasai** = East Africa (Kenya, Tanzania).
- Sweden and Finland have **Sami**, not Inuit.
- **Koryak** = Russia; **Kirghiz** = Central Asia.

UP LINK

UP's own tribes are few: **Tharu** (Terai: Lakhimpur Kheri, Bahraich), **Buksa**, **Gond**, **Kharwar** (Sonbhadra). The Tharu, like the Maasai and Bedouin, are a classic "tribe-region" pair.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Bushmen (San) = **Kalahari**, Botswana
2. Maasai = **Kenya**, Tanzania
3. Bedouin = **Arabia**; Kirghiz = Central Asia
4. Pygmies = **Congo**; hunter-gatherers
5. Eskimo/Inuit = Canada, Alaska, Greenland
6. Sami (Lapps) = Scandinavia
7. Semang = **Malaysia**; Ainu = **Japan**
8. Teda = Libya; Beja = Egypt; Lur = Iran
9. Koryak = Kamchatka
10. Kayak = hunting; umiak = transport

QUANTUM IAS & PCS

Chapter 17

World Agriculture and Livestock

Built from 65 UPPSC questions (1990–2023). Focus: leading producers of crops, crop-region matching, plantation crops, shifting cultivation names, milk.

ASKED MOST · REVISE FIRST

- **India** = top **milk** producer 5× 2003, 2008, 2010, 2011, 2012
- **Brazil** = top **coffee** producer; **Vietnam** second (and second exporter); estates called **fazendas** 5× 2009, 2012, 2013, 2016, 2017
- **Thailand** = top **natural rubber** producer (about one-third) 4× 2001, 2005, 2006, 2010
- **Brazil** = top **sugarcane**; India second; **Cuba** = "Sugar Bowl" 4× 2008, 2013, 2015, 2016
- **China** = top tea exporter (by value) 2× 2016, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

17.1 Cereals

- Leading producer of both wheat and rice: none now (India leads rice, China wheat) (*) 2× 2004, 2006
- India in rice: **first** (2023, 2024); second earlier 1× 2013
- Largest rice area: **India** 1× 2016
- Wheat, descending: **China, India, Russia, USA** 1× 2008
- Highest rice yield (of options): **China** 1× 2015
- Hybrid rice most popular in **China** 1× 2013
- World rice gene bank: **Philippines** (IRRI) 1× 2015
- Rice two-thirds of exports: none (Thailand nearest; India now top exporter) (*) 2× 2010, 2013
- Second in corn: **China** (USA first) 1× 2013
- Maize as staple: **Middle Africa** 1× 2007

17.2 Sugar, Fibres, Plantations

- Largest sugarcane producer: **Brazil**; second: **India** 2× 2008, 2016
- Top beet producers: Russia and France (later data reshuffle, *) 1× 2010
- Sugar from beet only: **Ukraine** 1× 2013
- Exports most of its sugar: **Mauritius** 1× 2011
- "Sugar Bowl": **Cuba** 2× 2013, 2015
- Highest cotton yield (2014): **Australia**; of options: **China** 2× 2015, 2013
- Largest coconut producer: **Indonesia** 1× 2008
- Largest rubber producer: **Thailand**; top two: **Thailand and Indonesia**; one-third from **Thailand** 4× 2001, 2006, 2005, 2010
- Top coffee producers: **Brazil and Vietnam**; top: **Brazil** 3× 2009, 2012, 2013
- Second coffee exporter: **Vietnam** 1× 2016
- Brazil's coffee estates: **fazendas** 1× 2017
- US coffee state: **Hawaii** 1× 2015
- Largest tea exporter: **China** 2× 2016, 2017
- Not a cocoa producer: **Latvia** 1× 2021
- Wrong: **coffee-Uzbekistan** 1× 2023
- Not correct: sugarcane is Malaysia's main plantation crop (it is **rubber, oil palm**) 1× 2008

17.3 Crop Regions

- **Coffee-Sao Paulo, jute-Ganges delta, rice-Yangtze, wheat-Prairies** 1× 2003
- **Coffee-Sao Paulo, rice-Yangtze, wheat-Huang He, tea-Kandy** 1× 2010
- **Cocoa-Ghana, coffee-Ivory Coast, tea-Kenya, sugarcane-South Africa** 1× 2013
- **Rubber-Malaysia, coffee-Colombia, olive-Italy, sugarcane-Mauritius** 2× 2010, 2013
- **China-iron ore, India-milk, Saudi Arabia-petroleum, Thailand-rubber** 1× 2000
- **Chad-petroleum, Niger-uranium, Rwanda-coffee, Benin-cotton** 1× 2017
- Producer statements: now China tobacco and Brazil soybean true (*) 1× 2000
- Coconut and sugarcane in the Philippines: **Spanish and Americans** 1× 2023
- Peanuts main crop of **Gambia** 2× 2012, 2013
- Groundnut native to **Brazil** 1× 2010

Crop	Leader
Rice / Wheat	India / China
Maize / Soybean	USA / Brazil
Sugarcane / Coffee	Brazil
Tea	China
Cocoa	Cote d'Ivoire
Rubber	Thailand
Coconut	Indonesia
Milk	India

17.4 Fruits and Vines

- India's rank in fruit: **second** 1× 2016
- Not tropical: **apple** 1× 1990
- Sacramento-San Joaquin valley: **California** 1× 1998

QUANTUM IAS & PCS

- Commercial viticulture is Mediterranean; 85% of grapes go to wine (R explains A) 1× 2005
- Citrus well developed in **Mediterranean regions** 1× 2020

17.5 Farming Systems

- Plantations clustered on coasts; Amazon lacks workers (2 and 3) 1× 1998
- Plantation crops: definitions differ (*) 1× 2004
- Crops between tree rows: **alley cropping** 1× 2007
- Monoculture: **commercial grain farming** 1× 2010
- Not shifting cultivation: **fazenda** 2× 2006, 2010
- **Jhum-SE Asia, Milpa-Mexico** correct; Ladang is not Sri Lanka (1 and 2) 1× 2023
- Blue-green algae biofertiliser: **rice** 1× 2010
- Not in tea: **morphine** 1× 2010
- First HYV seeds: **Mexico** (Borlaug, wheat) 1× 2008
- WTO Green Box: **research and plant protection** (1 and 4) 1× 2009

SHIFTING CULTIVATION NAMES



17.6 Narcotics and Livestock

- Heroin from **opium** 1× 2007
- Golden Crescent: **Afghanistan, Iran, Pakistan** 1× 2007
- Shahtoosh: **chiru** (Tibetan antelope) 1× 2006
- Hurt beef's image: **mad cow disease** 1× 2005
- Largest milk producer: **India** 4× 2003, 2008, 2010, 2012
- Milk order: **India, USA, China, Russia** (then) 1× 2011
- Highest milk per cow (of options): **Denmark** 1× 2018

TRAP ALERT

- **Fazenda** = coffee estate, not shifting cultivation; **Chena** = Sri Lanka.
- Tea exports: China by **value**, Kenya by **volume**.
- No one country now leads in both rice and wheat.

UP LINK

UP is India's top producer of **milk, wheat and sugarcane**; India itself leads the world in milk. Coffee and rubber are absent in UP, which grows mango, guava and potato.

LAST-MINUTE 10

1. India = milk, rice (now), second in fruit
2. China = wheat, tea export, hybrid rice
3. Brazil = sugarcane, coffee (fazendas); Vietnam second
4. Thailand = rubber; Indonesia = coconut
5. Cuba = Sugar Bowl; Ukraine = beet sugar
6. IRRI gene bank = Philippines; HYV = Mexico
7. Gambia = peanuts; groundnut native to Brazil
8. Milpa = Mexico; Ladang = Indonesia; Chena = Sri Lanka
9. Golden Crescent = Afghanistan, Iran, Pakistan
10. Shahtoosh = chiru

Chapter 18

World Minerals and Energy Resources

Built from 71 UPPSC questions (1991–2025). Focus: coalfield and iron-ore region matching, leading producers (which change over time), mining centres, oilfields and reserves, uranium.

ASKED MOST · REVISE FIRST

- **Coalfield matching**: **Ruhr, Saar-Germany; Donbas/Donetsk-Ukraine; Kuzbas-Russia; Lancashire, Bristol-UK; Appalachian-USA; Red Basin-China; NSW-Australia** 8× 1999, 2007, 2010, 2013, 2013, 2022, 2023, 2025
- **Mining centres**: **Mesabi-iron, Kalgoorlie-gold, Chuquicamata-copper, Kimberley-diamond, Witwatersrand-gold, Katanga-copper, Butte-silver, Kinta-tin** 6× 1991, 1999, 2004, 2010, 2017, 2020
- **Mexico** = top **silver**; **Chile** = top **copper** ore and **iodine** 6× 1993, 1996, 2011, 2014, 2018, 2022
- **Kazakhstan** = top **uranium** producer; **Australia** = largest reserves; **Canada** = Uranium City 5× 2006, 2009, 2009, 2010, 2014
- Largest oil reserves: **Venezuela** 3× 2010, 2012, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Coal

- All true: China leads coal; Donetsk-Ukraine; Saar-Germany; Appalachian-USA 1× 1999
- **Donbass**: **coal**; largest Ukrainian coal centre: **Donetsk** 2× 2007, 2023
- **Kuzbass**: coal and iron ore; **Novosibirsk** a centre (both) 1× 2023
- **Appalachian-USA, Lancashire-England, Ruhr-Germany, Kuzbas-Russia** 1× 2025
- **Kuzbas-Russia, Red Basin-China, Bristol-UK, New South Wales-Australia** 1× 2013
- **Donetsk-Ukraine, Kuznetsk-Russia, Lancashire-UK, Saar-Germany** 1× 2013
- **Ruhr**: **Germany** 2× 2010, 2022
- Over half the world's coal: **China** (USA per older key) 1× 2006
- **Rukwa** (Tanzania): **coal** 1× 2010

18.2 Iron Ore

- Producer order: **Australia, Brazil, China, India** now (*) 1× 2008
- Largest producer of options: **Brazil**; **India** among old options (*) 2× 2013, 1991
- France no longer a European iron producer (Lorraine mines closed 1997) (*) 2× 2010, 2013
- **Lorraine-France, Midlands-UK, Kiruna-Sweden, Kostanay-Kazakhstan** 1× 2010
- Wrong: **Germany-Normandy** (Normandy = France) 2× 2016, 2020
- **Mount Newman**: **iron ore** (Pilbara, Australia) 1× 2023
- Lake Superior region: **iron ore** (Mesabi) 1× 2017
- Great Lakes region industry: **steel and engineering** 1× 2016

Iron-ore region	Country
Pilbara (Mount Newman)	Australia
Carajas, Itabira	Brazil
Mesabi Range	USA
Kiruna	Sweden
Lorraine, Normandy	France
Krivoi Rog	Ukraine
Kostanay	Kazakhstan

18.3 Metals

- Largest copper ore producer: **Chile**; key **China** (refined copper) 2× 1996, 1998
- Chile leads in copper due to Andean **porphyry copper** (R explains A) 1× 2022
- **Chuquicamata**: **copper** 1× 2017
- Africa's top copper producer (then): **Zambia**; now **DR Congo** 1× 2010
- Japan nearly self-sufficient in **copper** (per key) 1× 2014
- Gold order: now **China, Russia, Australia** (*) 1× 2008
- **Kalgoorlie**: **gold**; **Coolgardie**: **Western Australia** 2× 1991, 2011
- Largest gold exporter to India: **Switzerland** 1× 2013
- ■ Largest silver producer: **Mexico** 3× 1993, 2011, 2014
- Aluminium among options: **India** (2023 data) 1× 1993
- Bauxite: **Australia** per key; **Guinea** now leads 2× 2013, 2015
- Tin: **Indonesia** (then); China now 1× 2010
- **Kinta Valley** (Malaysia): **tin** 1× 2020
- Titanium: **Russia** (per key) 1× 2009
- **Postmasburg** (South Africa): **manganese** 1× 2020
- Diamond: **Russia** 2× 2012, 2013
- Iodine: **Chile** 1× 2018
- Afghanistan's $1 trillion find: **lithium** 1× 2008
- Wrong: **copper-cinnabar** (cinnabar = mercury ore) 1× 2017
- All pairs broadly correct (tin-Bolivia, iron-Brazil, silver-Mexico, nitrate-Peru) (*) 2× 2010, 2013
- **Silver-Mexico, mica-India, iron ore-Russia, copper-Chile** 1× 2017

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18.4 Mineral Regions

- **Coal-Karaganda, gold-Highveld, iron-Krivoi Rog, oil-San Joaquin Valley** 1× 2010
- Wrong: **Mesabi-copper** (Mesabi = iron) 1× 2004
- **Iron-Mesabi, petroleum-Baku, copper-Bingham, uranium-Port Radium** 1× 1999
- **Baku**: **petroleum** 1× 1993
- Correct: **copper-Arizona** 1× 2007
- Wrong: **Saar-iron ore** (Saar = coal) 1× 2007
- **Copper-Katanga, diamond-Kimberley, gold-Witwatersrand, silver-Butte** 1× 2010

LEADERS THAT CHANGED

Iron ore
China/India (old keys) to Australia

↓

Gold
South Africa to China (since 2007)

↓

Bauxite
Australia to Guinea (2022-23)

↓

Petroleum
Saudi Arabia/Russia to USA (since 2018)

↓

Copper in Africa
Zambia to DR Congo

18.5 Petroleum and Gas

- Producer order: now **USA, Russia, Saudi Arabia** (*) 3× 1998, 2008, one-liner
- Largest producer (key, then): **Russia** 1× 2010
- Middle East order: **Saudi Arabia, Iran, Iraq, Kuwait** 1× 2008
- West Asia's top producer: **Saudi Arabia** 1× 2017
- South-East Asia's top producer: **Indonesia** 1× 2004
- Largest reserves: **Venezuela**; **Iran** among older options 4× 2010, 2012, 2013, 2002
- All true: Middle East reserves, Alaska, USA producer-importer 1× 1998
- Alaska statements: both correct 1× 2021
- Wrong: **Kuwait-Kashagan** (Kashagan = **Kazakhstan**) 1× 2024
- **Pegu Yoma** (Myanmar): **mineral oil** 1× 2022
- Largest gasohol producer and user: **USA** 1× 2011
- "Peace Pipeline" (Iran-Pakistan-India): none of the options 1× 2007
- Top crude supplier to India 2017-18: **Iraq** 1× 2017

Oilfield	Country
Ghawar	Saudi Arabia
Burgan	Kuwait
Kirkuk, Rumaila	Iraq
Kashagan, Tengiz	Kazakhstan
Baku	Azerbaijan
Prudhoe Bay	Alaska
Maracaibo	Venezuela
Pegu Yoma	Myanmar

18.6 Uranium and Others

- Top uranium producer: **Kazakhstan** 2× 2009, 2010
- Largest reserves: **Australia** 1× 2009
- **Uranium City**: **Canada**; important uranium ore: **Canada** 2× 2006, 2014
- Not a major Russian resource (per key): **uranium** 1× 2017
- Natural camphor: from a tree native to **China and Japan** 1× 2014

TRAP ALERT

- **Mesabi** = iron; **Saar** = coal; **cinnabar** = mercury; **Kashagan** = Kazakhstan.
- "Largest producer" keys age fast; check the year.
- Chile leads **mine** copper; China leads **refined** copper.

UP LINK

UP's minerals are minor (Sonbhadra coal, limestone, Lalitpur rock phosphate); the **Singrauli** coalfield straddling UP-MP is India's equivalent of a Ruhr-style energy hub.

LAST-MINUTE 10

1. Ruhr, Saar = Germany; Donbas = Ukraine; Kuzbas = Russia
2. Karaganda = Kazakhstan; Red Basin = China
3. Mesabi = iron (USA); Kiruna = Sweden; Krivoi Rog = Ukraine
4. Mount Newman (Pilbara) = iron, Australia
5. Chile = copper, iodine; Mexico = silver
6. Kalgoorlie = gold; Kimberley = diamond; Witwatersrand = gold
7. Katanga = copper, cobalt; Kinta = tin
8. Kazakhstan = uranium; Canada = Uranium City
9. Venezuela = largest oil reserves; USA top producer now
10. Kashagan = Kazakhstan; Baku = oil

Chapter 19

Industrial Cities and Regions of the World

Built from 25 UPPSC questions (1990–2022). Focus: city-industry matching, shipbuilding traps, industrial regions, footloose industry.

ASKED MOST · REVISE FIRST

- **City-industry matching**: **Pittsburgh-steel, Detroit-automobiles, Osaka-cotton, Dundee-jute, Sheffield-cutlery, Glasgow and St. Petersburg-ships, Anshan-steel, Havana-cigars, Abadan-oil refinery** 6× 2002, 2005, 2005, 2010, 2011, 2022
- Wrong shipbuilding pairs: **Birmingham, Chelyabinsk, Moscow, Venice**; Camden is New Jersey, not Florida 5× 1996, 1998, 2001, 2005, 2016
- **Footloose** industry: **software, electronics** 2× 2013, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

19.1 Iron and Steel

- **Hamilton-Canada, Birmingham-UK, Essen-Germany, Anshan-China** 1× 2011
- Wrong: **Birmingham-shipbuilding** 1× 1998
- Wrong: **Chelyabinsk-shipbuilding** 1× 2001
- Wrong: **Moscow-shipbuilding** 1× 2005
- Wrong: **Mesabi-coalfield** (iron ore) 1× 2013
- Market-based steel locations: **Japan** 2× 2010, 2013

19.2 Textiles, Automobiles, Ships

- **Pittsburgh-iron and steel, Shanghai-cotton textiles, Dundee-jute, Leningrad-shipbuilding** 1× 2005
- **Nagoya-cotton textiles, Salem-iron and steel, Los Angeles-aircraft, Abadan-oil refinery** 1× 2010
- **Osaka-cotton, Detroit-automobiles, Cuba-cigars, St. Petersburg-shipbuilding** 1× 2022
- **Anshan-iron and steel, Detroit-automobiles, Havana-cigars, Nagoya-cotton** 1× 2002
- Cotton textile centre: **Osaka** 1× 1990
- Wrong: **Venice-shipbuilding** (per key) 1× 1996
- **Anshan-steel, Glasgow-shipbuilding, Ottawa-paper, Sheffield-cutlery** 1× 2005
- Wrong: **Houston-automobiles** (oil, space) 1× 2010
- Wrong: **Camden-Florida** (New Jersey) 1× 2016
- Wrong: **cotton textiles-Detroit** 1× 2015

City	Industry
Pittsburgh, Gary	Steel (USA)
Detroit	Automobiles
Houston	Oil, space
Sheffield / Birmingham	Cutlery / Steel (UK)
Glasgow, Belfast	Ships
Dundee / Manchester	Jute / Cotton
Essen (Ruhr)	Steel (Germany)
Magnitogorsk, Chelyabinsk	Steel (Urals)
St. Petersburg	Ships
Osaka / Nagoya	Cotton / Cars, textiles
Anshan	Steel (China)
Abadan / Havana	Refinery / Cigars

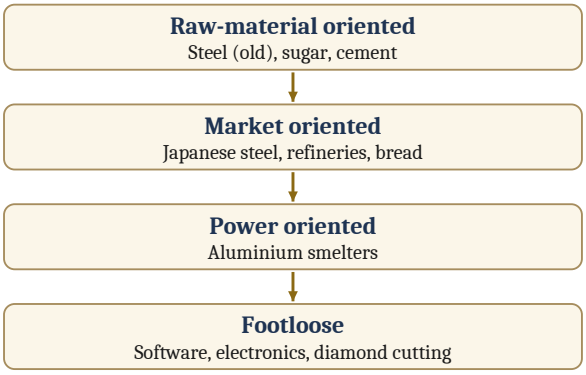
19.3 Industrial Regions

- **Kinki-Japan, Canton-China, Lorraine-France, Belo Horizonte-Brazil** 1× 2016
- Correct: **Ruhr-Germany** 1× 2016
- Wrong: **Scotland industrial region-Sweden** (UK) 1× 2013

19.4 Textiles and Location

- Biggest wool producer (key): **China** 1× 2013
- German wool centre: **Wuppertal** 1× 2013
- Leading cotton textile producer: **China** 1× 2005
- Footloose: **software** 1× 2013
- Raw material not primary: **electronics** 1× 2015
- Japan leads through **high technology** 1× 2008

HOW INDUSTRIES CHOOSE A SITE



TRAP ALERT

- **Birmingham** = steel; **Glasgow, Belfast** = ships.
- Moscow and Chelyabinsk are inland: **no shipbuilding**.
- **Camden** = New Jersey; **Houston** = oil, not cars.

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UP LINK

UP's footloose-industry hubs are **Noida** (electronics, mobiles, software) and Lucknow; Kanpur is UP's "Manchester", like Osaka in Japan and Manchester in the UK.

LAST-MINUTE 10

1. Pittsburgh = steel; Detroit = cars; Houston = oil
2. Sheffield = cutlery; Glasgow = ships; Dundee = jute
3. Essen = steel; Wuppertal = wool
4. St. Petersburg = ships; Chelyabinsk = steel
5. Osaka = cotton (Manchester of Japan); Nagoya = cars, textiles
6. Anshan = steel; Havana = cigars; Abadan = refinery
7. Kinki = Japan; Canton = China; Belo Horizonte = Brazil
8. Japan steel = market-based
9. Footloose = software, electronics
10. Camden = New Jersey

QUANTUM IAS & PCS

Chapter 20

World Transport and Miscellaneous

Built from 41 UPPSC questions (1991–2024). Focus: sea routes, railways and tunnels, ports, geographers and their ideas, HQs of organisations, assorted facts.

ASKED MOST · REVISE FIRST

- **North Atlantic** = busiest sea route; links two industrial zones 4× 1998, 2002, 2003, 2004
- **Bermuda Triangle** = **North Atlantic** 5× 2001, 2002, 2003, 2008, 2015
- Busiest port by cargo: **Shanghai**; Japan's busiest: **Yokohama** 3× 2004, 2010, 2015
- Geographers: **Taylor-neo-determinism, Sauer-planting hearths, Engler-floral regions, Lee-migration, Spencer and Thomas-11 cultural worlds** 4× 1994, 2016, 2017, 2024

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

20.1 Routes, Railways, Tunnels

- North Atlantic busiest; joins two industrial zones (R explains A) 2× 2002, 2003
- Busiest sea route: **North Atlantic** 2× 1998, 2004
- Not on the Trans-Siberian: **Sochi** 1× 2019
- Rail length: **USA**, China, Russia, India, Canada (no code fits, *) 2× 2008, 2014
- Longest rock tunnel: Delaware Aqueduct, USA (none of options, *) 1× 2013
- **Seikan** rail tunnel: **Japan** 1× 2012

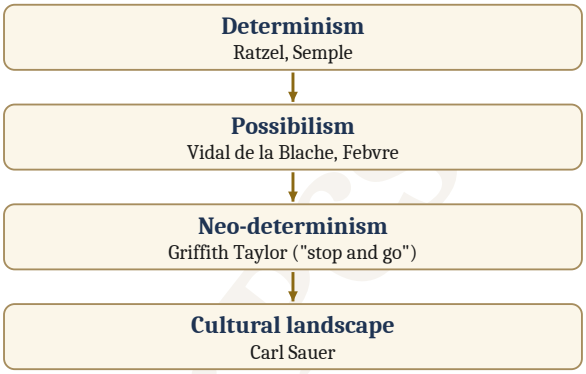
20.2 Ports and Shipbuilding

- Busiest seaport by cargo: **Shanghai**; second biggest (of options): **Shanghai** 2× 2010, 2004
- Japan's busiest port: **Yokohama** 1× 2015
- **Alexandria**: **Egypt** 1× 2007
- **Port Diamond**: **South Africa** 1× 2010
- **Gwadar**: **Balochistan** 1× 2021
- Wrong: **Igarka-China** (Igarka = Russia, Yenisei) 1× 2016
- Largest shipbuilder by value: **China** 1× 2010

20.3 Geographers and Concepts

- **Planting hearths-C.O. Sauer, cryophyte steppe-Arctic tundra, world floral regions-Adolf Engler, Puszta-Hungary** 1× 2024
- **Neo-determinism**: **Griffith Taylor** 1× 1994
- Spencer and Thomas: **11** cultural worlds 1× 2016
- Migration theory: **Lee** 1× 2017
- **Umland**: the **area surrounding a town** 1× 2017
- **Isogonic** lines: equal **magnetic declination** 1× 1998

SCHOOLS OF THOUGHT



20.4 Places and Facts

- **Grand Banks**: **eastern coast of North America** (Newfoundland) 1× 2010
- ■ **Bermuda Triangle**: **North Atlantic** 4× 2001, 2002, 2003, 2008
- **Golden Triangle-opium (SE Asia), Bermuda Triangle-North Atlantic, rice bowl of the Far East-Myanmar, Red Basin-China** 1× 2015
- Correct: **Morocco-Atlas** 1× 2002
- Wrong: "Queen of the Adriatic"-**Rome** (it is **Venice**) 1× 2015
- "Country of 40 tribes": **Kyrgyzstan** 1× 2015
- True: Cuba sugar bowl, Hong Kong SAR, Australia federal (1, 2, 4); USA not top milk producer 1× 2005
- Two wrong: Bosnia not Baltic; Nova Scotia = Canada (*) 1× 2013
- Not true: species diversity **increases** towards the poles (it decreases) 1× 2019
- Indian Ocean bases due to geopolitical value (key b; source a) 1× 2019

20.5 Organisations, Energy, History

- **IMO**: **London** 1× 2016
- Wrong: **FAO-New York** (FAO = **Rome**) 1× 2002
- Not in OPEC: **China** 1× 2002
- Nuclear plant count order: USA, France, Japan, Russia then (*) 1× 2013
- 99% energy from hydro: **Norway** 1× 2021
- **Chernobyl**: **Ukraine** 1× 2013
- Great Leap Forward famine: **China** 1× 2013
- US launch centre: **Cape Kennedy** (Canaveral) 1× 1991
- Largest wine producer: **Italy** 1× 2010
- Asia's biggest gold market (then): **China** 1× 2017
- Tree age: **counting rings** 1× 2006

Body	HQ
IMO	London
FAO, WFP, IFAD	Rome
WHO, WTO, ILO	Geneva
ICJ	The Hague
UNESCO, OECD	Paris
IAEA, OPEC	Vienna
UNEP	Nairobi
ISA	Gurugram

TRAP ALERT

- **FAO** = Rome, not New York; **IMO** = London.
- **Igarka** = Russia; **Gwadar** = Balochistan (Pakistan).
- Species diversity is highest at the **Equator**.
- Queen of the Adriatic = **Venice**.

UP LINK

UP links: the **Golden Quadrilateral** and **NW-1 (Prayagraj-Haldia)** are India's busiest land and inland water routes; **Jewar** airport and the **Eastern Dedicated Freight Corridor** put UP on new trade routes.

LAST-MINUTE 10

1. North Atlantic = busiest sea route
2. Shanghai = busiest port; Yokohama = Japan's busiest
3. Seikan tunnel = Japan; Sochi not on Trans-Siberian
4. Gwadar = Balochistan; Igarka = Russia
5. Taylor = neo-determinism; Sauer = planting hearths
6. Engler = floral regions; Lee = migration
7. Bermuda Triangle = N. Atlantic; Golden Triangle = opium
8. Kyrgyzstan = 40 tribes; Venice = Queen of the Adriatic
9. IMO = London; FAO = Rome
10. Norway = 99% hydro; Chernobyl = Ukraine

QUANTUM IAS & PCS